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Future-Complete Quotients

Recursive Distinction

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Contents

Abstract

A finite source calculus is developed from one formal primitive: a distinction has a complete family of resultants, and every resultant may distinguish again. Finite unlabelled non-plane hierarchies retain containment, multiplicity, and complete resultant families. Complete recursive views and complete one-occurrence continuation decks reconstruct every finite source hierarchy. The central closure theorem characterizes exactly when a quotient reporter supports an autonomous continuation law: the complete reported response must be constant on every reporter fibre. Failure induces a least response-forced refinement; success implies stable blindness to every distinction erased by the quotient. The calculus further constructs least closed combinations of reporters, finite predictive quotients, target-free expansion systems, and common-source relations between incomparable perspectives. An occurrence-sensitive prime triad and its nine-state, thirty-six-relation realization are added as a concrete quotient-closure test.

Keywords: recursive distinction; predictive quotient; behavioural closure; finite reconstruction; reporter refinement

Scope and status: The reconstruction, closure, refinement, finite quotient, and common-source results are theorem-level statements in the declared finite constructor. Interpretations beyond that constructor are stated separately.

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1 Introduction

Any formal account of distinction begins after at least one distinction has already been made. Terms such as *state*, *event*, *set*, and *operation* belong on the reportable side of that boundary. The first task is therefore to mark where formal description begins without representing the absence of description as one more object.

We use P_0/R for that boundary. It denotes no reportable distinction and is assigned no carrier, point, value, state, law, temporal order, modality, or truth value. This is a constraint on the theory's language rather than a claim about a hidden substrate. Every symbol in the paper is already a distinction, so every proof and computation belongs to P_1, the first reportable perspective, whose primitive separation is binary.

The sole source rule is

$$\mathsf{D}: \quad \text{a distinction has resultants, and every resultant may distinguish.}$$

Completeness and recursion specify how the primitive is reported. Completeness forbids the omission of a resultant; recursion permits the same rule to act at any resultant. Neither introduces a clock, probability, coordinate system, objective, or update schedule. A complete q-resultant family written in P_1 is consequently an encoding of q retained occurrences. When $q > 2$, indexing that record by a hypothetical P_{q-1} does not grant access to a native nonbinary logic.

The least finite transcription that keeps containment, multiplicity, and complete resultant families visible is an unlabelled, non-plane rooted hierarchy. A record is either an unresolved terminal $\bullet$, or a containing occurrence with an unordered multiset of at least two child records. Equal child descriptions may repeat: descriptive equality does not cancel occurrence. The hierarchy is a declared representation within P_1, not an ontological identification of R with a tree.

Given a source hierarchy and a reporter Q that identifies some source states, when does an operation on the source induce a law on the reported states? The closure criterion answers this directly: two states with the same report must have the same complete reported response. Otherwise the reporter has merged states with incompatible futures. Retaining the incompatibility, rather than choosing or averaging an outcome, determines the least refinement needed to make the report lawful.

Closure also fixes a limit. Once a law descends to a quotient, all of its iterates remain functions of that quotient. The law cannot reconstruct a distinction that its own state space consistently erased. Thus autonomy and blindness are not separate phenomena in this calculus; both are consequences of the same fibre condition.

The development proceeds from the finite hierarchy and its reconstructive views to complete output decks, quotient closure, forced refinement, and closed reporter combination. Later sections introduce finite interaction systems, a target-free expansion process, and common-source comparisons. These latter constructions carry additional hypotheses and are kept separate from the primitive core.

2 Principal results and scope

The central theorem may be stated without interpretive terminology:

> **Complete-behaviour closure.** Let Q report the declared finite hierarchies, whose complete occurrence decks are finite. A single-valued continuation law descends to the Q-states if and only if complete reported response is constant on every fibre of Q. If constancy fails, equality through all finite response depths defines the least finer reporter on which every reported arity closes.

The paper establishes five groups of results.

1. **Source structure and reconstruction.** The declared hierarchy grammar transcribes complete recursive distinction. Within it, the containing occurrence is the unique unselected site at which one retained role persists through a distinction without deletion, duplication, or child selection. Complete recursive views reconstruct a hierarchy at finite depth, as does the deck formed by replacing every terminal occurrence once. Counts and flattened arity profiles are strictly coarser because they lose shared containment.
2. **Law, refinement, and stable loss.** The closure criterion is necessary and sufficient. Its failure determines a least response-forced refinement; its satisfaction proves stable blindness, since response iteration preserves the reporter's fibres. Grouping-depth reporters form a closed hierarchy, and the least closed combination of two reporters is obtained by closing their direct pair. This combination is associative and commutative up to fibres, with $X \vee_c X \simeq Cl_\infty(X)$.
3. **Finite effect and interaction quotients.** On a declared finite carrier, an effect family induces the least same-referent quotient sufficient for those effects. For codomains with at least two values, properly finer reports admit strictly more compatible candidate laws. For $|X| = q \geq 2$ and a pointed operation of arity $m \geq 1$, finite interaction contexts generate the coarsest focus-preserving predictive quotient and stabilise after at most $q - 2$ strict refinements.
4. **Test systems.** A target-free binary expansion process yields local replacement, name invariance, commutation at separate current occurrences, ancestry-restricted histories, exact continuation factorisation, and nontermination. A second construction separates histories, ordered forms, order-free forms, and availability, making the permanence of closed information loss explicit.
5. **Relations among perspectives.** For two quotients of a common source, direct translation exists exactly along refinement. Incomparable quotients are related through the source-induced relation; their joint quotient is the least report retaining both distinction families and may be strictly finer than either marginal.

3 Complete distinction and behavioural closure

3.1 Formal boundary and provenance

The calculus begins with a restriction: its first inscription must not be mistaken for a description of what precedes inscription. We therefore separate the formal boundary from every object constructed beyond it.

3.1.1 3.1.1 The P_0/R boundary

Boundary statement. P_0/R denotes no reportable distinction. It is assigned no carrier or internal relation and is not identified with a familiar mathematical object. Any such identification would already distinguish that representation from alternatives, and so would belong on the reportable side of the boundary.

The notation is eliminative, not descriptive: it marks the limit of positive formal attribution. Even the sentence "there is no reportable distinction" is written from a reportable perspective. It fixes the provenance of the calculus rather than supplying a positive theory of R.

3.1.2 The first reportable perspective

Perspective convention. P_1 is the first reportable perspective: one distinction with two resultants. Every word, equation, diagram, proof, and program in this paper is consequently a P_1 inscription. The calculus cannot step outside that representational condition to certify the native experience or mathematics of another perspective.

A binary medium may nevertheless encode a finite family with $q > 2$ distinguishable occurrences. Accordingly, "a q-resultant distinction" means precisely:

> P_1 records, without deleting an occurrence, a complete family of q distinguishable resultants.

This is an external P_1 presentation, not a claim of access to a native nonbinary perspective.

Figure 1 records this asymmetry: the boundary is not a source node, while binary and higher-arity records are constructions on its reportable side.

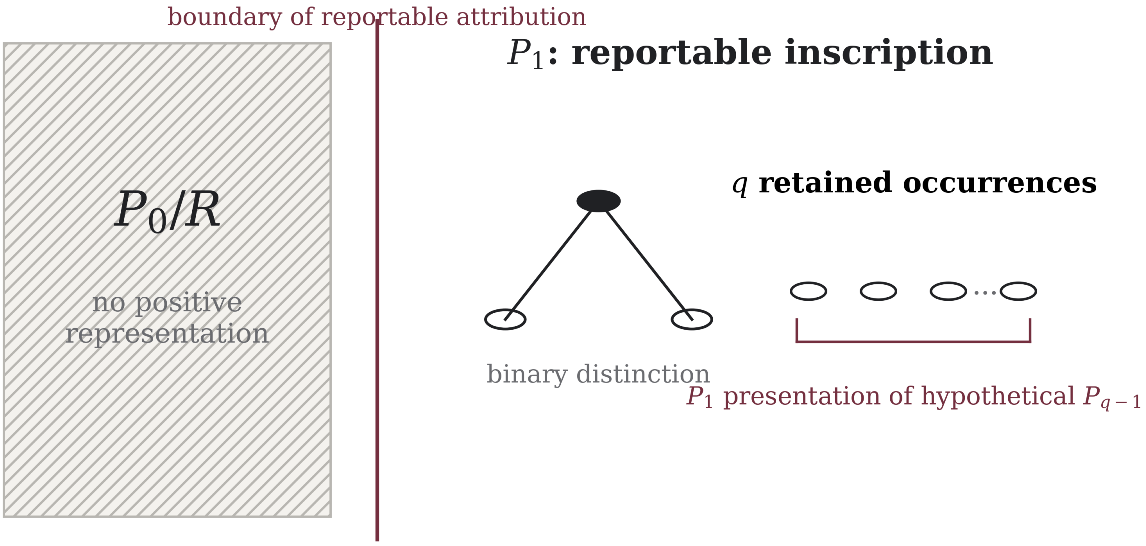

Figure 1. Representation boundary and perspective indexing. The vertical rule marks the limit of positive attribution; no transition is drawn across it.

3.1.3 The sole source rule

The construction retains one source rule:

$$\boxed{\mathsf{D}: \quad \text{a distinction has resultants, and every resultant may distinguish.}}$$

Completeness means that no resultant is discarded from the report; recursion means that the same rule is applied again to a resultant. Neither adds a second source rule. In particular, D contains no scheduler, coordinates, probability law, metric, conservation equation, logical valuation, or target phenomenon.

Definitions below specify the mathematical constructor; numbered propositions, theorems, and corollaries state consequences of their displayed hypotheses. Appendix A records their provenance.

3.2 The minimal complete hierarchy

3.2.1 Grammar and equality

Definition 3.2.1 (finite complete hierarchy). Let $\mathcal{H}$ be generated by

$$T ::= \bullet \quad \text{or} \quad \langle T_1, \ldots, T_q \rangle, \qquad q \geq 2.$$

The symbol $\bullet$ records a resultant with no further reported distinction. The term $\langle T_1, \ldots, T_q \rangle$ records a containing occurrence and all of its q present resultants. Children are unordered, since D supplies no order; equal descriptions retain their occurrence multiplicity rather than being identified.

Equality in $\mathcal{H}$ is therefore isomorphism of finite unlabelled, non-plane, rooted hierarchies with multiplicity. A node of the underlying rooted hierarchy is called an *occurrence*. An occurrence is *terminal* when it is represented by $\bullet$, and *internal* when it carries a nonempty child family.

This grammar is minimal relative to four declared reporting obligations:

1. a distinction leaves one containing occurrence at which it occurred;

2. every present resultant is retained;
3. every resultant may recursively carry the same form of record;
4. no child order or label is inserted without another supplied distinction.

"Minimal" is relative to these four obligations. The claim is not that R is a hierarchy, but that $\mathcal{H}$ is the least explicit P_1 transcription used here that records recursive containment and the complete resultant family without introducing labels or order.

For $T \in \mathcal{H}$, let $L(T)$ be its number of terminal occurrences and $h(T)$ the maximum number of containment edges from the root to a terminal. For an internal occurrence v, let $q(v) \geq 2$ denote its arity. These are measurements of a completed record, not source primitives.

We use finite multisets throughout. The notation $\{\{a_1, \ldots, a_m\}\}$ retains the number of occurrences of each value; multiset equality includes multiplicity. If $f : A \to B$ and M is a finite multiset on A, then f_*M denotes its multiplicity-preserving pushforward.

3.2.2 The information order on reporters

Definition 3.2.2 (reporter and refinement). A reporter on $\mathcal{H}$ is any function $Q : \mathcal{H} \to X$. For reporters X and Y, write

$$X \preccurlyeq Y$$

when equality of Y-reports implies equality of X-reports. Equivalently, X factors through Y. Thus Y retains every distinction retained by X, and perhaps more. Reporters are identified *up to fibres*: $X \simeq Y$ means $X(T) = X(U)$ if and only if $Y(T) = Y(U)$ for all $T, U \in \mathcal{H}$.

The order compares retained distinctions only; it carries no independent ranking by truth, observer quality, physical resolution, or temporal progress.

3.2.3 Why the containing occurrence persists

Suppose an occurrence carries one focal role x. After it distinguishes into a complete unlabelled family, the old role could apparently be erased, assigned to one child, copied to every child, or retained at the containing occurrence. The first choice violates retention; the second introduces an unsupplied child selection; the third turns one role into q. Only the fourth preserves exactly one focal role.

Proposition 3.2.3 (unselected persistence). Inside the declared hierarchy transcription, the containing occurrence is the unique placement that retains one old focal role through its distinction without deleting, selecting, or duplicating that role.

This is an identity principle of the transcription. It assigns no objecthood or persistence to R: the children form the complete new resultant family, with no default heir to the old role.

3.2.4 First invariants and the cost of flattening

The completed record supports exact accounting identities that are absent from the source rule.

Proposition 3.2.4 (balance). Every $T \in \mathcal{H}$ satisfies

$$\boxed{L(T) = 1 + \sum_{v \text{ internal}} (q(v) - 1)\,.}$$

Proof. The unresolved record $\bullet$ has one terminal and no internal contribution. Replacing one terminal by a family of q resultants increases the terminal total by $q-1$ and adds one internal occurrence contributing exactly $q-1$ to the displayed sum. Every finite hierarchy is obtained by finitely many such replacements, so induction on the number of internal occurrences proves the identity. □

For a homogeneous q-resultant hierarchy with I internal occurrences, the balance reduces to

$$L = 1 + (q-1)I.$$

For homogeneous binary hierarchies this becomes $L = I + 1$. Both identities belong to the finite P_1 transcription.

Now put $e_v = q(v) - 1$. Then

$$L(T) - 1 = \sum_v e_v.$$

This scalar factors through two distinct losses:

$$\mathcal{H} \to \{\{e_v\}\} \to \sum_v e_v = L - 1.$$

The first arrow erases which internal occurrences contain which others. The second erases even the multiset decomposition of the total excess. To see the first loss directly, let $T_{a,b}$ have a children at the root and b children below each root child. Reversing the two layers yields $T_{b,a}$. For $a \neq b$ their containment histories differ, but both have ab terminals. The equality $ab = ba$ is therefore visible at the scalar shadow while the two complete hierarchies remain distinct.

Flattening can lose information even when terminal total, root arity, and the total number of grandchildren are all retained. Consider

$$A = \langle \bullet, \langle \bullet, \bullet \rangle, \langle \bullet, \bullet, \bullet, \bullet \rangle \rangle$$

and

$$B = \langle \bullet, \langle \bullet, \bullet, \bullet \rangle, \langle \bullet, \bullet, \bullet \rangle \rangle.$$

Both have seven terminals, three root children, and six grandchildren, but their complete nested child-family reports are respectively

$$\{\{\varnothing, 2[*], 4[*]\}\} \quad \text{and} \quad \{\{\varnothing, 3[*], 3[*]\}\}.$$

Hence a flattened occurrence union does not determine recursive nesting. An associativity law on a flat algebra may be perfectly sound, but the quotient has changed the object under study: scalar and flat reporters are proper shadows of the hierarchy.

3.3 Complete present-resultant views

Two complete families will be used: the children present at an occurrence, and the possible one-occurrence continuations of a terminal. They are different constructions.

3.3.1 Present-family readout

Definition 3.3.1 (present-resultant operator). For any expression e evaluable on hierarchies, define

$$[\![\Delta e]\!](\bullet) = \varnothing,$$

and

$$[\![\Delta e]\!]\left(\langle T_1, \ldots, T_q \rangle\right) = \{\{[\![e]\!]\left(T_1\right), \ldots, [\![e]\!]\left(T_q\right)\}\}.$$

Thus Δ returns the entire present child multiset. It neither selects nor labels a child, and it performs no ordering, averaging, or scalarization.

3.3.2 Recursive self-view

Begin with one undifferentiated focal report

$$V_0(T) = *.$$

Apply the same report recursively to every present resultant:

$$V_{r+1}(\bullet) = (*, \varnothing),$$

$$V_{r+1}\left(\langle T_1, \ldots, T_q \rangle\right) = \left(*, \{\{V_r\left(T_1\right), \ldots, V_r\left(T_q\right)\}\}\right).$$

Write

$$V_\omega(T) := \left(V_r(T)\right)_{r \geq 0},$$

so equality of V_ω-reports means equality at every finite view depth.

The first coordinate retains the focal role at its containing occurrence. The second records the absence or presence of a child family and, when present, applies the same reading to every child occurrence.

Theorem 3.3.2 (finite reconstruction). If $T \in \mathcal{H}$ has height h, then $V_{h+1}(T)$ determines T. Consequently, equality through every finite view depth has exactly the equality fibres of $\mathcal{H}$:

$$\boxed{V_{h(T)+1}(T) \text{ reconstructs } T, \qquad V_\omega \simeq \mathcal{H}.}$$

Proof. We induct on the height. If $T = \bullet$, then $V_1(T) = (*, \varnothing)$, which identifies the terminal record. Let $T = \langle T_1, \ldots, T_q \rangle$ have height $h > 0$. Its V_{h+1}-report contains, with multiplicity, the family $\{\{V_h\left(T_1\right), \ldots, V_h\left(T_q\right)\}\}$. Every child has height at most $h - 1$; by the induction hypothesis, its V_h-report reconstructs it. Reconstruct every member of the complete child multiset and place that multiset under the single containing occurrence. This recovers T. Because every finite T has finite height, equality at all finite view depths implies equality in $\mathcal{H}$, while equality in $\mathcal{H}$ plainly implies equality of every view. □

Theorem 3.3.3 (strict view hierarchy). Erasing the deepest retained family from a V_{r+1}-report yields its V_r-report, so

$$V_0 \preccurlyeq V_1 \preccurlyeq V_2 \preccurlyeq \cdots .$$

Each refinement is strict on the unbounded family of finite hierarchies.

Proof. The forgetful map is immediate from the recursive definition. A terminal and an internal occurrence agree under V_0 and differ under V_1. More generally, if T and U first differ at view depth $r+1$, embed each beneath the same newly supplied containing context; their first difference is shifted to depth $r+2$. Hence examples witnessing strictness exist at every finite level. □

Here r counts nested applications of the P_1 readout. It has no assigned interpretation as time, distance, physical scale, knowledge, computational complexity, or perspective index.

Reconstruction requires neither names nor child order. Complete multiplicity and recursive containment suffice. Present-resultant access is therefore faithful on finite hierarchies, whereas scalar count identifies many nonisomorphic records.

3.4 Complete one-occurrence future decks

3.4.1 Definition and elementary invariants

Fix an externally reported arity $q \geq 2$.

Definition 3.4.1 (complete occurrence deck). For $T \in \mathcal{H}$, let $E_q(T)$ be the finite multiset obtained by replacing each terminal occurrence of T, one occurrence at a time, by a q-resultant occurrence $\langle \bullet, \ldots, \bullet \rangle$. If symmetric terminal occurrences yield isomorphic outputs, the output hierarchy appears with the corresponding multiplicity.

The deck is an exhaustive family of one-occurrence continuations, not a probability distribution. Each output $U \in E_q(T)$ satisfies

$$L(U) = L(T) + q - 1,$$

and the total occurrence multiplicity of the deck is

$$\left|E_q(T)\right|_{occ} = L(T).$$

Where Δ reads the current children of one focal occurrence, E_q enumerates every terminal occurrence at which another distinction can be recorded.

3.4.2 Source reconstruction from one complete deck

Theorem 3.4.2 (deck reconstruction). For every fixed integer $q \geq 2$ and all finite $T, U \in \mathcal{H}$,

$$\boxed{E_q(T) = E_q(U) \Rightarrow T = U.}$$

Proof. Let $h = h(T)$. Expanding a terminal at depth h creates an output of height $h+1$, while expanding any shallower terminal creates an output of height at most h. Hence the maximum height among members of $E_q(T)$ is $h+1$, and every maximum-height member results from expansion at a maximum-depth terminal.

Choose a maximum-height output W. In the source T, no internal occurrence can lie at depth h, since it would have a terminal descendant below depth h, contradicting maximality of h. Therefore W has exactly one internal occurrence at depth h: the newly inserted q-resultant occurrence. Contract that occurrence to a terminal. The contraction uniquely recovers T. Thus a complete deck determines its source. Equal decks have a common maximum-height member and reconstruct the same source, proving $T = U$. □

Thus the unquotiented one-step continuation family is faithful to the full finite hierarchy, not merely to terminal count. Before passing to quotients, the following subsection records the exact locality available when two replacements occur on opposite sides of a retained cut. Section 3.5 then asks which parts of this continuation structure survive a quotient report.

3.4.3 Persistent cuts and restricted two-step interaction

Occurrence provenance permits an exact cut without choosing a child or flattening either side. For a finite hierarchy T, let $\Lambda(T)$ be its set of terminal occurrences. For a marked occurrence v, write $D = T|_v$ for the subhierarchy rooted at v and $T = C[D]$ for the associated one-hole context. The terminals below v form $\Lambda_v(T) \subseteq \Lambda(T)$, canonically identified with $\Lambda(D)$. Write

$$\Lambda_v^{out}(T) = \{u \in \Lambda(T) : u \notin \Lambda_v(T)\}$$

for the terminal occurrences outside the marked subhierarchy. Let

$$B_q = \langle \bullet, \ldots, \bullet \rangle, \qquad q \geq 2,$$

where the bracket contains exactly q terminal entries, and write $T\left[u \leftarrow B_q\right]$ for replacement of the terminal occurrence u by B_q. Define the occurrence-indexed multisets

$$E_q^{in}(T; v) = \{\{T\left[u \leftarrow B_q\right] : u \in \Lambda_v(T)\}\},$$

$$E_q^{out}(T; v) = \{\{T\left[u \leftarrow B_q\right] : u \in \Lambda_v^{out}(T)\}\}.$$

Outputs that become isomorphic are aggregated only after their originating terminal occurrences have been counted.

Theorem 3.4.3 (persistent cut and exact reunion). For every finite $T \in \mathcal{H}$, every marked occurrence v, and every integer $q \geq 2$,

$$\boxed{E_q(T) = E_q^{in}(T; v) \uplus E_q^{out}(T; v).}$$

Moreover,

$$E_q^{in}(T; v) = \{\{C[W] : W \in E_q(D)\}\},$$

with multiplicity preserved. Every member of the inside deck retains the outer context $C[-]$, and every member of the outside deck retains the marked subhierarchy D unchanged.

Proof. Every terminal occurrence of T lies in exactly one of the disjoint sets $\Lambda_v(T)$ and $\Lambda_v^{out}(T)$. The definition of $E_q(T)$ indexes one replacement by each terminal occurrence, so this set partition induces the displayed multiset disjoint union, including all multiplicities. Under the canonical identification $\Lambda_v(T) \cong \Lambda(D)$, replacing u below v changes only D and yields $C\left[D\left[u \leftarrow B_q\right]\right]$; this proves the inside-deck formula. A replacement outside $\Lambda_v(T)$ is disjoint from the marked occurrence and therefore leaves D unchanged. □

The cut also isolates an exact, deliberately local order-independence result.

Theorem 3.4.4 (opposite-side commutation). Let $T = C[D]$ and v be as above. Fix integers $q, r \geq 2$. For every $u \in \Lambda_v(T)$ and $w \in \Lambda_v^{out}(T)$, both terminal occurrences survive when the other is replaced, and

$$\boxed{T\left[u \leftarrow B_q\right][w \leftarrow B_r] = T\,[w \leftarrow B_r]\left[u \leftarrow B_q\right].}$$

Consequently, the complete occurrence-indexed two-step multiset obtained by a q-replacement inside the cut followed by an r-replacement outside is equal to the multiset obtained in the opposite order.

Proof. The terminal occurrences u and w are distinct and lie in disjoint occurrence regions. Replacing u changes only the subhierarchy at u, so w remains a terminal occurrence with the same provenance; symmetrically, replacing w leaves u available. Both composites are therefore the simultaneous hierarchy obtained from T by inserting B_q at u and B_r at w. Pairing the two orderings by the same occurrence pair (u, w) is a bijection of their finite indexing sets and preserves the resulting hierarchy. Aggregating isomorphic outputs thus preserves equal multiplicities. □

The restriction to opposite sides of a retained cut is essential. Identical or nested occurrences require a separate interaction analysis. This result supplies a concrete two-step law at source level; the quotient criterion below decides whether any reporter retains enough information for that law, or any complete continuation law, to descend.

3.5 Autonomous quotient laws

A reporter may retain less than the full hierarchy. The pertinent question is whether its reported state determines its complete reported behaviour.

Let $Q : \mathcal{H} \to X$ be any declared reporter. For a deck $E_q(T)$, define its pushed report

$$Q_* E_q(T) = \{\{Q(W) : W \in E_q(T)\}\},$$

with occurrence multiplicity retained.

Definition 3.5.1 (autonomous q-law). An autonomous continuation law for Q at arity q is a function

$$F_{Q,q} : Q\,(\mathcal{H}) \to MFin\,(Q\,(\mathcal{H}))$$

such that

$$F_{Q,q}\,(Q(T)) = Q_* E_q(T)$$

for every $T \in \mathcal{H}$. Here $MFin(X)$ denotes finite multisets on X.

Thus an autonomous law is internal to the reporter: the present Q-state determines the complete multiset of next Q-states without reference to a hidden representative in $\mathcal{H}$.

Theorem 3.5.2 (autonomy/closure criterion). An autonomous q-law exists if and only if the complete pushed deck is constant on every Q-fibre:

$$\boxed{Q(T) = Q(U) \Rightarrow Q_*E_q(T) = Q_*E_q(U) \quad \text{for all } T, U \in \mathcal{H}.}$$

Proof. If $F_{Q,q}$ exists and $Q(T) = Q(U) = x$, then

$$Q_*E_q(T) = F_{Q,q}(x) = Q_*E_q(U),$$

so the condition is necessary. Conversely, suppose the condition holds. For $x \in Q(\mathcal{H})$, choose any T with $Q(T) = x$ and define $F_{Q,q}(x) = Q_*E_q(T)$. Fibre constancy makes this definition independent of the representative, so it is a function satisfying the required equation. □

Figure 2 separates the two cases of Theorem 3.5.2: descent when complete decks agree on every fibre, and refinement when they do not.

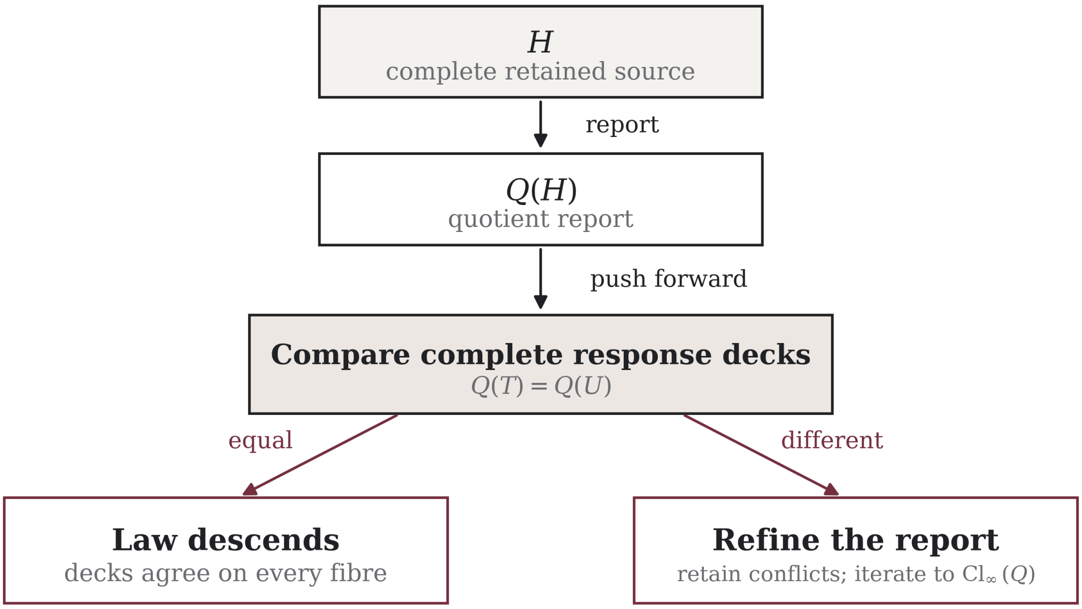

Figure 2. Complete-behaviour closure. Fibre constancy yields an autonomous quotient law; conflicting decks force the least refinement that retains every behaviourally relevant distinction.

We call Q *q-closed* when it satisfies the criterion, and *all-arity closed* when it is q-closed for every integer $q \geq 2$.

If closure fails, the complete response at a fibre is the set of all decks it supports:

$$\widehat{F}_{Q,q}(x) = \{Q_*E_q(T) : Q(T) = x\}.$$

This set-valued family is not an autonomous single-valued law. Its plurality records one precise fact: Q has erased a distinction on which its complete response depends. No probabilistic or stochastic interpretation follows without additional structure.

Corollary 3.5.3 (count forced by complete multiplicity). Every q-closed reporter Q refines terminal count L.

Proof. If $Q(T) = Q(U)$, closure gives equal pushed decks. Equal finite multisets have equal total multiplicity. These multiplicities are respectively $L(T)$ and $L(U)$, hence $L(T) = L(U)$. □

Terminal count is itself all-arity closed, because its complete law is

$$L \mapsto L \text{ occurrences of } L + q - 1.$$

Therefore L is the coarsest reporter closed under complete terminal replacement with occurrence multiplicity. The result is operation-relative: count is not sufficient for Δ, whose recursive views reconstruct the full hierarchy.

The identity reporter is closed. So is the multiset of arity excesses $\{\{q(v) - 1\}\}$: a q-replacement appends $q - 1$, while the current excess sum fixes deck multiplicity by Proposition 3.2.4. Internal-node count I, by contrast, is not closed when multiplicity is retained. A binary star and a ternary star both have $I = 1$, but their decks contain two and three occurrences, respectively. Its first response refinement is exactly (I, L), which is closed: every replacement sends (I, L) to $(I + 1, L + q - 1)$ with multiplicity L.

Accordingly, a reporter may already be autonomous, may acquire finitely many missing coordinates, or may require an unbounded refinement tower. The criterion decides which case occurs from the reported behaviour itself.

3.6 Response-forced completion

When closure fails, the conflicting responses identify which fibres must split. Iterating this observation gives a canonical completion without prescribing target classes.

3.6.1 Iterated response reporters

Fix a reporter $Q : \mathcal{H} \to X$. Define

$$R_0^Q(T) = Q(T).$$

Recursively, let $R_{r+1}^Q(T)$ retain $R_r^Q(T)$ together with, for every integer $q \geq 2$, the complete multiplicity deck of R_r^Q-reports of members of $E_q(T)$:

$$R_{r+1}^Q(T) = \left(R_r^Q(T), (R_{r*}^Q E_q(T))_{q \geq 2}\right).$$

Only equality fibres matter; no canonical code for the arity-indexed tuple is needed. Let $T \equiv_r^Q U$ mean $R_r^Q(T) = R_r^Q(U)$. Since R_{r+1}^Q retains R_r^Q,

$$\equiv_{r+1}^Q \subseteq \equiv_r^Q .$$

Define the limiting equivalence

$$T \equiv_\infty^Q U \quad \Leftrightarrow \quad T \equiv_r^Q U \text{ for every finite } r.$$

Write $Cl_\infty(Q)$ for the quotient reporter whose fibres are the $\equiv_\infty^Q$-classes.

3.6.2 Least-closure theorem

Theorem 3.6.1 (least all-response closed refinement). On the finite hierarchy family $\mathcal{H}$, $Cl_\infty(Q)$ is all-arity closed. Moreover, it is the least discriminating all-arity closed reporter that refines Q. Explicitly:

1. $Q \preccurlyeq Cl_\infty(Q)$;
2. $Cl_\infty(Q)$ is q-closed for every $q \geq 2$;
3. if S is all-arity closed and $Q \preccurlyeq S$, then $Cl_\infty(Q) \preccurlyeq S$.

Proof. The first statement follows because $R_0^Q = Q$ is retained at every later depth.

For closure, suppose $T \equiv_\infty^Q U$ and fix $q \geq 2$. The occurrence decks $E_q(T)$ and $E_q(U)$ are finite. For each $r \geq 0$, equality of $R_{r+1}^Q(T)$ and $R_{r+1}^Q(U)$ implies equality of the complete R_r^Q-decks. Hence there exists a multiplicity-preserving bijection between the occurrence-indexed members of $E_q(T)$ and $E_q(U)$ that pairs outputs having equal R_r^Q-reports.

Let Π_r be the finite set of all such bijections. Each Π_r is nonempty. Since equality at depth $r+1$ implies equality at depth r, the sets descend:

$$\Pi_0 \supseteq \Pi_1 \supseteq \Pi_2 \supseteq \cdots .$$

They are subsets of the one finite set of all bijections between the two finite occurrence decks. A descending sequence of nonempty subsets of a finite set has nonempty intersection. Choose $\pi \in \bigcap_r \Pi_r$. Every output is paired by π with an output equal under every finite R_r^Q, hence equal under $\equiv_\infty^Q$. The pushed $Cl_\infty(Q)$-decks are therefore equal. Since q was arbitrary, the limiting reporter is all-arity closed.

For leastness, let S be any all-arity closed reporter refining Q, and suppose $S(T) = S(U)$. We prove by induction on r that $R_r^Q(T) = R_r^Q(U)$. The base case follows from $Q \preccurlyeq S$. Assume the assertion holds at depth r for every pair of equal S-reports. Because S is q-closed, equality $S(T) = S(U)$ gives, for every $q \geq 2$, a multiplicity-preserving pairing of $E_q(T)$ and $E_q(U)$ whose paired outputs have equal S-reports. By the induction hypothesis those paired outputs have equal R_r^Q-reports. Hence the complete R_r^Q-decks agree for every q, and $R_{r+1}^Q(T) = R_{r+1}^Q(U)$. Induction yields equality at all finite response depths, so $T \equiv_\infty^Q U$. Thus equality under S implies equality under $Cl_\infty(Q)$, which is exactly $Cl_\infty(Q) \preccurlyeq S$. □

The universal property is relative to the stated carrier and operation: finite hierarchies, complete one-terminal replacement at every reported arity, and multiplicity-preserving decks.

Restricting the constructor to a fixed $q \geq 2$ gives, by the same proof, the least q-closed refinement $Cl_q(Q)$. A reporter can be closed for one replacement arity and fail for another; the all-arity completion is the common refinement obtained when every reported arity is retained.

In the preceding closure proof, finiteness is used when the descending family of compatible pairings is shown to have a common member. Although $\mathcal{H}$ is infinite, no uniform finite stabilisation depth is asserted. Infinite occurrence decks would require an additional compactness or choice principle.

Two states separate at the first response depth where their complete behaviour differs, and later separations occur only when subsequent responses reveal another conflict. State refinement and autonomous law are therefore determined together.

3.7 Stable blindness

3.7.1 The ceiling theorem

Response completion repairs unstable loss. It cannot recover an erasure already compatible with every supported response.

Theorem 3.7.1 (stable blindness). If Q is all-arity closed, then for every finite $r \geq 0$,

$$\boxed{R_r^Q(T) = R_r^Q(U) \quad \Leftrightarrow \quad Q(T) = Q(U).}$$

Consequently, $Cl_\infty(Q) \simeq Q$.

Proof. Because R_r^Q retains Q, equality of R_r^Q-reports implies equality of Q-reports. For the converse, proceed by induction. It is immediate for $r = 0$. Suppose R_r^Q is a function of Q. Since Q is closed, $Q(T) = Q(U)$

implies that for each $q \geq 2$ the occurrence decks $E_q(T)$ and $E_q(U)$ can be paired with equal Q-reports. Applying the inductive functional dependence to paired outputs gives equal R_r^Q-reports, so the complete R_r^Q-decks agree. Together with equality of the retained first coordinate, this gives $R_{r+1}^Q(T) = R_{r+1}^Q(U)$. Thus every R_r^Q is both a function of Q and a refinement retaining Q, so their fibres agree. □

Erased distinctions may remain in $\mathcal{H}$; the theorem says that iteration of the autonomous Q-law supplies no behavioural witness for them. Unequal responses expose unstable erasure, while closure makes the erasure stable under the observed operation.

This yields a sharp ceiling. The views V_r reconstruct because Δ retains recursive child grouping. A closed scalar or flat reporter may erase that relation while retaining a perfectly autonomous replacement law; further iteration then cannot restore it.

Stable blindness is thus an invariant of a reporter–operation pair: distinctions inside a closed fibre are unavailable to that reporter under that operation.

3.7.2 A strict hierarchy of closed grouping resolutions

Grouping can still be retained by a richer reporter; it cannot be regenerated by a closed reporter that has erased it. The following family makes the distinction explicit.

Let $G_0(T) = L(T)$. For $r \geq 0$, put

$$G_{r+1}(\bullet) = \bullet_r$$

and

$$G_{r+1}\left(\langle T_1, \ldots, T_q\rangle\right) = \langle G_r\left(T_1\right), \ldots, G_r\left(T_q\right)\rangle_{unlabelled}.$$

Write $G_\omega(T) := (G_r(T))_{r\geq 0}$; equality under G_ω therefore means equality at every finite grouping depth.

The terminal mark closes the transcription at the chosen depth. Each G_r is all-arity closed: a G_{r+1} state retains the full child multiset at level G_r, and one terminal replacement changes exactly one child occurrence according to the complete G_r law. Equal parent outputs are then aggregated with multiplicity.

Proposition 3.7.2 (grouping tower). The reporters form a strict chain on the unbounded finite hierarchy family:

$$\boxed{L = G_0 \prec G_1 \prec G_2 \prec \cdots \prec G_\omega \simeq \mathcal{H}.}$$

For a hierarchy of height h, G_h determines it. No fixed finite G_r determines every finite hierarchy.

Proof. Erasing the outermost retained grouping layer maps G_{r+1} to G_r. The reconstruction induction used for V_r shows that depth h suffices for a height-h hierarchy. Strictness follows by taking any pair first separated at depth $r+1$ and placing both beneath one additional common container. The difference moves one layer deeper and survives at the next resolution while disappearing at the previous one. □

If $\pi_r : G_{r+1} \to G_r$ erases one layer and $F_{r,q}(s)$ denotes the complete G_r output deck from state s, the laws commute with projection:

$$\boxed{(\pi_r)_* F_{r+1,q}(s) = F_{r,q}\left(\pi_r(s)\right).}$$

The tower therefore consists of compatible autonomous shadows, each with its own stable blindness. Full finite reconstruction requires progressively richer retained relations, not repeated iteration of one coarse closed law.

Corollary 3.7.3 (closed ceiling). If $Q \preccurlyeq S$ and S is all-arity closed, then

$$Cl_\infty(Q) \preccurlyeq S.$$

Proof. The leastness clause of Theorem 3.6.1 applies directly: S is a closed refinement of Q, so the least closed refinement of Q cannot retain more distinctions than S. □

Hence any family of reporters factoring through the same closed reporter remains confined to its fibres under response iteration. Recovering a lost relation requires a richer constructor that retains it.

3.8 Combining perspectives and forced relations

The direct pair of two autonomous reporters need not be autonomous. Each coordinate may preserve a marginal while both erase their same-occurrence relation.

3.8.1 The pairing obstruction

Let $D(T)$ be the multiset of terminal depths in T, and let $A(T)$ be the multiset of the final internal arities appearing immediately above terminal occurrences. For the root-terminal record $\bullet$, use a distinguished null mark $\perp$ in A; its first replacement removes $\perp$ and inserts q copies of q. With that boundary convention, both reporters are defined on all of $\mathcal{H}$ and are closed under E_q. Selecting an occurrence with values (d, a) changes the depth marginal by

$$D \mapsto D - \{d\} + q\{d + 1\},$$

and the final-arity marginal by

$$A \mapsto A - \{a\} + q\{q\}.$$

However, the separate multisets do not retain which d was paired with which a at a terminal. There are finite hierarchies T_1, T_2 with identical marginals

$$D = (2, 2, 2, 3, 3, 3, 3, 4, 4), \qquad A = (2, 2, 2, 2, 2, 3, 3, 3, 3),$$

but different same-occurrence profiles:

Terminal pair (d, a)	**Multiplicity in** C_1	**Multiplicity in** C_2
$(2, 2)$	2	1
$(2, 3)$	1	2
$(3, 2)$	1	2
$(3, 3)$	3	2
$(4, 2)$	2	2

For a compact exact witness, put

$$B = \langle \bullet, \bullet \rangle, \qquad C_3 = \langle \bullet, \bullet, \bullet \rangle, \qquad J = \langle \bullet, B \rangle,$$

$$E = \langle \bullet, C_3, J \rangle, \qquad F = \langle \bullet, \bullet, B \rangle.$$

Then

$$T_1 = \langle B, E \rangle, \qquad T_2 = \langle F, \langle \bullet, F \rangle \rangle.$$

The nine terminal paths give the same two marginals but the displayed distinct joint profiles. Their direct-pair decks differ because replacement acts on an occurrence carrying both values. Replacing (d, a) removes that occurrence and inserts q copies of $(d + 1, q)$, so output multiplicities depend on C, not on D and A separately. Thus

$$D \text{ closed and } A \text{ closed} \not\Rightarrow (D, A) \text{ closed.}$$

The first response refinement restores the missing relation

$$C(T) = \{\{(d, a) \text{ at each terminal occurrence}\}\}.$$

The reporter C is closed, since replacement sends (d, a) to q copies of $(d + 1, q)$. To see that the first response refinement is exactly C, fix any $q \geq 2$. For fixed marginals D, A, the two update maps

$$d \mapsto D - \{d\} + q\{d + 1\}, \qquad a \mapsto A - \{a\} + q\{q\}$$

are injective. Equality of two images in the first map would imply $-e_d + qe_{d+1} = -e_{d\prime} + qe_{d\prime+1}$, whose coefficient at the least index on which d and $d\prime$ differ is impossible unless $d = d\prime$; the second map is injective because equality implies $e_a = e_{a\prime}$. Each member of the pushed direct-pair deck therefore identifies the selected pair (d, a), and deck multiplicity recovers its multiplicity in C. Conversely C determines both marginals and every pushed deck. Hence

$$R_1^{(D,A)} \simeq C.$$

3.8.2 Least closed combination

For arbitrary reporters X and Y, let (X, Y) be their direct-pair reporter and define

$$\boxed{X \vee_c Y := Cl_\infty(X, Y).}$$

Theorem 3.8.1 (universal property of closed combination). The reporter $X \vee_c Y$ is the least all-arity closed common refinement of X and Y. Precisely:

1. $X \preccurlyeq X \vee_c Y$ and $Y \preccurlyeq X \vee_c Y$;
2. $X \vee_c Y$ is all-arity closed;
3. if S is all-arity closed with $X \preccurlyeq S$ and $Y \preccurlyeq S$, then $X \vee_c Y \preccurlyeq S$.

Proof. The direct pair retains both coordinates, so $X \preccurlyeq (X, Y)$ and $Y \preccurlyeq (X, Y)$; response completion then gives $(X, Y) \preccurlyeq Cl_\infty(X, Y)$. Closure and leastness follow from Theorem 3.6.1 applied to the direct pair. □

Corollary 3.8.2 (combination laws). Up to equality fibres,

$$X \vee_c Y \simeq Y \vee_c X,$$

$$(X \vee_c Y) \vee_c Z \simeq X \vee_c (Y \vee_c Z),$$

and

$$X \vee_c X \simeq Cl_\infty(X).$$

In particular, if X is already closed, then $X \vee_c X \simeq X$.

Proof. Coordinate exchange leaves the set of closed common refinements unchanged, proving commutativity. Both parenthesizations in the second equation are the least closed reporter refining X, Y, Z, so the universal property makes them fibre-equivalent. Repeating X adds no initial distinction beyond X, and closure then gives $Cl_\infty(X)$; for closed X, stable blindness reduces this to X. □

Up to fibres, closed reporters therefore form an idempotent, commutative, associative combination structure under $\vee_c$. The algebra is derived from behavioural completion.

The results in this section concern the finite hierarchy carrier and the specified complete operations. Report depth, occurrence, containment, closure, and multiplicity receive no physical interpretation without an additional empirical bridge.

4 Generated perspectives, predictive closure, and target-free continuation

The source boundary and the finite hierarchy now being fixed, this part studies structures generated by recursive distinction. Sets, maps, partitions, addresses, and algorithms remain reports made within P_1. For $r \geq 2$, a construction retaining r resultants may be indexed P_{r-1}; when $r > 2$, that index records arity without claiming native access by P_1 to the hypothetical perspective it names.

Provenance is stated once at the opening of each construction. General theorems carry the hypotheses under which they are proved; finite censuses are reported separately from those proofs. This keeps the formal question narrow: once an interaction has been declared or generated, which distinctions and laws does it force?

4.1 Same-referent quotients and the reversal of constraint

4.1.1 Retained indistinction

Conditional construction. Let C be a context carrier and let

$$a_i : C \to X_i, \qquad 1 \leq i \leq h,$$

be effects already aligned on the same contexts. The alignment is a premise: the construction does not infer that two unrelated carriers concern the same referent. Write $A = (a_1, \ldots, a_h)$, and define

$$c \sim_A d \quad \Leftrightarrow \quad a_i(c) = a_i(d) \text{ for every } i.$$

The quotient

$$Q_A = C/{\sim_A}, \qquad q_A : C \to Q_A,$$

retains exactly the distinctions jointly made by A. Each coordinate factors as $a_i = \bar{a}_i \circ q_A$. Resultant renaming leaves the quotient unchanged, while a relabelling of C merely relabels its blocks. Thus the quotient is a structure of distinctions, not a structure of the temporary names used to encode them.

Theorem SRQ (same-referent shadow criterion). For any candidate effect $t : C \to Y$, the following are equivalent:

1. there is a map $\bar{t} : Q_A \to Y$ such that $t = \bar{t} \circ q_A$;
2. t is constant on every A-indistinction class:

$$c \sim_A d \Rightarrow t(c) = t(d).$$

When it exists, $\bar{t}$ is unique.

Proof. Factorisation immediately implies constancy on fibres. Conversely set $\bar{t}([c]_A) = t(c)$. Fibre constancy makes this independent of the representative, and surjectivity of q_A makes it unique. □

The criterion separates three cases. A **shadow** factors through Q_A and adds no distinction, although it may merge quotient classes. A **faithful presentation** has the same indistinction relation as Q_A; on its reached image it is only a relabelling. A **new distinction** is witnessed by some $c \sim_A d$ with $t(c) \neq t(d)$. Adjoining such a t changes the generated referent.

The quotient is also universal. If $r : C \to S$ is sufficient for every retained effect, so that $a_i = \alpha_i \circ r$, then $r(c) = r(d)$ implies $c \sim_A d$. A unique map $\rho : r(C) \to Q_A$ therefore satisfies $q_A = \rho \circ r$. Thus Q_A is the least-distinguishing sufficient carrier: a sufficient report may retain more, but it must map to Q_A.

4.1.2 More distinction forces less

For a partition Q of C and a resultant set Y, define its admissible shadow family

$$Sh_Y(Q) = \{t : C \to Y : t \text{ is constant on each } Q\text{-block}\}.$$

Theorem CR (constraint reversal). If Q_2 refines Q_1, then

$$Sh_Y(Q_1) \subseteq Sh_Y(Q_2).$$

If the refinement is proper and $|Y| \geq 2$, the inclusion is strict.

Proof. Constancy on a Q_1-block entails constancy on every Q_2-subblock. If Q_2 properly splits a Q_1-block, assign two distinct resultants to two of its Q_2-subblocks while keeping each subblock constant. The resulting effect descends through Q_2 but not through Q_1. □

If Q has b blocks and $|Y| = q < \infty$, then $|Sh_Y(Q)| = q^b$.

This is the precise sense in which more distinction forces less. A finer base admits more candidate effects and therefore determines less about which candidate is realised. A one-block finite carrier admits only the q constant effects, whereas the discrete partition admits all $q^{|C|}$ effects. The one-block case is still a P_1 representation of a finite audit carrier; it is not a representation of P_0/R.

Two identification barriers follow immediately. First, **self-seeding** is circular: if t is used to construct $Q_{A\cup\{t\}}$, then t necessarily factors through that quotient. Second, a **discrete base** is non-predictive: every candidate factors through it. A meaningful same-referent test must therefore generate Q_A independently of the candidate and must retain some indistinction.

4.1.3 Reached laws and their completions

Let

$$e : C \to U = X_1 \times \cdots \times X_h, \qquad e(c) = (a_1(c), \ldots, a_h(c)),$$

and let $b : C \to Y$ be a target coordinate observed on the same contexts.

Theorem RLD (reached-law descent). There is a unique law on reached configurations,

$$\phi : e(C) \to Y, \qquad b = \phi \circ e,$$

if and only if $e(c) = e(d) \Rightarrow b(c) = b(d)$.

Proof. If $b = \phi \circ e$, equal e-values have equal b-values. Conversely define $\phi\,(e(c)) = b(c)$. The displayed constancy condition makes this independent of the chosen representative, and every element of $e(C)$ has a representative; hence ϕ exists and is unique. □

This is not yet a law on all of U. If U and Y are finite, with $M = |U|$, $r = |e(C)|$, and $q = |Y|$, then the number of complete extensions $\Phi : U \to Y$ realizing b is exactly zero when descent fails and exactly q^{M-r} when descent holds.

Assume descent holds. For $|Y| \geq 2$, the resulting partial law has a unique complete extension precisely when every configuration is reached. Completion and identification are different operations: retaining every extension guarantees that unreached assignments remain available, but does not choose among them. When $|Y| = 1$, descent is automatic and the extension is uniquely constant even with unreached configurations; a singleton codomain supplies no identifying constraint.

The same reasoning yields an exact forcing threshold. Fix a source coordinate i, a source focus x, and a target focus y, and let $H_{i,x} = \{u \in U : u_i = x\}$. Assume descent holds and $|Y| \geq 2$. Every compatible completion sends all of $H_{i,x}$ to y if and only if the full face is reached and every reached member of the face has that target output. If $m = \left|\{u \in H_{i,x} : u \notin e(C)\}\right|$ and no reached member contradicts y, exactly q^{M-r-m} of the q^{M-r} compatible completions preserve the face. An unreached point is not negligible: it is precisely a retained degree of freedom.

Formally, SRQ is a kernel-factorisation criterion of the kind used in universal algebra, while RLD is a reached-state analogue of the sufficient-state condition in automata theory (Burris and Sankappanavar 1981; Nerode 1958). The difference is one of provenance: no machine semantics is supplied in advance. The retained effects generate the quotient, and the unreached configurations quantify what those effects leave unresolved.

4.2 Interaction-generated predictive perspectives

The same-referent quotient presupposes an effect family. We now derive a focus-preserving quotient from interaction itself, before naming any downstream target. Depending on the law, the generated quotient may be discrete or may retain nontrivial indistinction.

4.2.1 Constructor and refinement

Declared inputs; conditional construction. Fix integers $q \geq 2$ and $m \geq 1$, a finite set X with $|X| = q$, and an internally pointed law fibre

$$(f, x), \qquad f : X^m \to X, \quad x \in X.$$

The q elements of X are the retained resultants. The carrier, arity, total law, and focus are declared. From them, the following construction is forced by interaction; it receives no metric, clock, arithmetic coordinate, desired partition, or comparison law.

For coordinate j and fixed values in all other coordinates, law application produces an elementary one-place continuation

$$\tau_{j,\mathbf{a}}(u) = f\left(a_1, \ldots, a_{j-1}, u, a_{j+1}, \ldots, a_m\right).$$

Let T_f be the monoid generated by these continuations and the identity. Begin with the intrinsic pointed distinction

$$u\,E_0\,v \quad\Leftrightarrow\quad (u = x \Leftrightarrow v = x).$$

For any equivalence E, define

$$u\,\mathcal{R}_f(E)\,v \quad\Leftrightarrow\quad u\,E\,v \text{ and } \tau(u)\,E\,\tau(v) \text{ for every elementary } \tau,$$

and iterate $E_{r+1} = \mathcal{R}_f(E_r)$. A block splits only when application of the retained law distinguishes two members relative to a distinction already generated. There is no scoring function and no target partition.

> **Theorem IPQ-1 (finite stabilisation).** For the finite carrier $|X| = q \geq 2$ and arity $m \geq 1$ fixed above, the chain

$$E_0 \succcurlyeq E_1 \succcurlyeq \cdots$$

stabilises after at most $q - 2$ strict refinements.

Proof. A strict step increases the number of blocks by at least one and no step merges blocks. The initial partition has two blocks and a partition of X has at most q. □

The bound is sharp. For the unary law $d_q(j) = \max(0, j-1)$ focused at 0, successive rounds isolate 1, then 2, and so on. The generated block counts are $2, 3, \ldots, q$, requiring exactly $q - 2$ strict steps.

> **Theorem IPQ-2 (all-finite-continuation characterisation).** At stabilisation,

$$u\,E_\infty\,v \quad\Leftrightarrow\quad (t(u) = x \Leftrightarrow t(v) = x) \text{ for every } t \in T_f.$$

Proof. Let $T_f^{\leq r}$ be the continuation words of length at most r, including the identity word. We prove by induction that

$$u\,E_r\,v \quad\Leftrightarrow\quad (t(u) = x \Leftrightarrow t(v) = x) \text{ for every } t \in T_f^{\leq r}.$$

For $r = 0$ this is the definition of E_0. Suppose it holds at r. By definition, $u\,E_{r+1}\,v$ precisely when $u\,E_r\,v$ and $\tau(u)\,E_r\,\tau(v)$ for every elementary continuation τ. The induction hypothesis applies to the first clause for words of length at most r and to the second for all words $t\tau$ of length at most $r+1$. This proves the induction step. Because the finite chain stabilises, choose N with $E_N = E_{N+1} = E_\infty$. Fixed-point equality then gives $E_{N+j} = E_\infty$ for every $j \geq 0$, so the displayed bounded-word characterisation holds at every length. Hence E_∞ is exactly agreement under all finite continuation words. □

4.2.2 Generated law and universal minimality

> **Theorem IPQ-3 (congruence and quotient law).** The stable equivalence is compatible with f:

$$u_i\,E_\infty\,v_i \qquad (1 \leq i \leq m) \quad\Rightarrow\quad f(u_1, \ldots, u_m)\;E_\infty\;f(v_1, \ldots, v_m)\,.$$

Proof. For $0 \leq j \leq m$, put

$$w_j = f\left(v_1, \ldots, v_j, u_{j+1}, \ldots, u_m\right).$$

Fix $j < m$ and hold every coordinate except the $(j+1)$st at the values displayed in w_j and w_{j+1}. This defines an elementary continuation τ. Since $u_{j+1}\, E_\infty\, v_{j+1}$ and E_∞ is a fixed point of $\mathcal{R}_f$, we have $\tau\left(u_{j+1}\right)\, E_\infty\, \tau\left(v_{j+1}\right)$, or $w_j\, E_\infty\, w_{j+1}$. Transitivity along $w_0, \ldots, w_m$ proves the claim. □

Therefore $Q_f = X/E_\infty$ carries a unique induced law

$$\bar{f}\left([u_1], \ldots, [u_m]\right) = \left[f\left(u_1, \ldots, u_m\right)\right]$$

with the exact commuting equation

$$\pi_f \circ f = \bar{f} \circ \left(\pi_f\right)^m.$$

Theorem IPQ-4 (coarsest predictive quotient). Let C be any equivalence on X for which $\{x\}$ is a class and C is a congruence of f. Then C refines E_∞.

Proof. Every such C refines E_0. If it refines E_r, congruence under each elementary continuation implies that it refines E_{r+1}. Induction gives refinement of the limit; IPQ-3 shows the limit itself is admissible. □

Thus Q_f is the unique least-distinguishing carrier that preserves the focus and supports the law. Coarser identification destroys either focus separation or law descent; finer presentation retains distinctions not forced by the interaction. Mathematically, E_∞ is the behavioural equivalence induced by all finite focus-reachability tests, and its computation is a specialised partition-refinement procedure (Moore 1956; Hopcroft 1971; Paige and Tarjan 1987; Rutten 2000). What is specific here is the observation family: it is generated internally from the pointed law before any external target is named.

Figure 3 shows the construction from the pointed law to its stable, focus-preserving quotient.

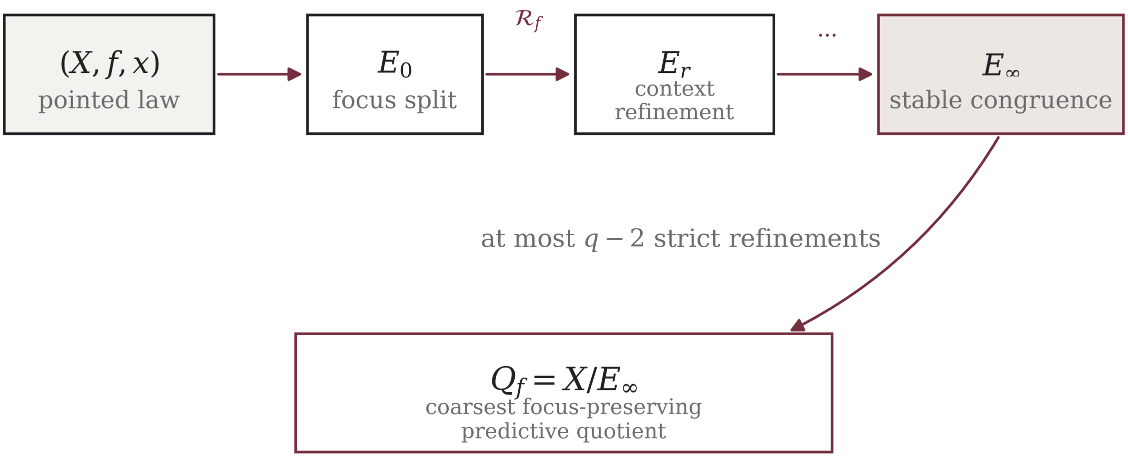

Figure 3. Interaction-generated predictive quotient. Repeated refinement separates only pairs distinguished by a finite continuation relative to the retained focus. The stable quotient carries the induced law.

At round r, the generated state is the partition E_r, not its block count. A scalar report may be taken only afterward:

$$k_r = |X/E_r| - 1.$$

For the sharp family this shadow is $(1, 2, \ldots, q-1)$. The ordering is logical dependence under repeated interaction, not physical time. If Q_f has s resultants, the corresponding P_1 report may be indexed P_{s-1}. Both k_r and the perspective index are read from the closed construction; neither was an input.

Finite audit. Every one of the $3^{3^2} = 19,683$ binary-input laws on a ternary carrier was checked at all three foci, giving $59,049$ pointed laws. The audit covered monotonicity, stabilisation, congruence, the universal property, quotient reconstruction, cyclic-renaming equivariance, and canonical signatures.

Generated quotient	Pointed laws
2 resultants (P_1 report)	3,825
3 resultants (P_2 report)	55,224
Total	59,049

These produce 9,238 label-free predictive isomorphism classes. The iterative and all-finite-continuation definitions agree in 113 complete small-law comparisons; all 486 unary ternary pointed permutation cases are equivariant. The delayed-focus depth formula was also checked for carrier sizes 2 through 10. These computations audit the theorems; they do not replace their proofs.

4.3 Comparing perspectives through a common source

Cross-perspective comparison requires a common source. Appending coordinates to a P_1 carrier and declaring $X \times Q$ a new perspective is not neutral: the product already contains a deterministic projection to X, so incomparable descriptions have been ruled out by construction.

4.3.1 Future-complete source quotients

Conditional source calculus. Let M be a finite family of source effects closed under declared primitive continuations, and let O_θ be a retained observation. Define

$$m\theta n \quad \Leftrightarrow \quad O_\theta(wm) = O_\theta(wn) \text{ for every finite primitive word } w.$$

The perspective $P_\theta = M/\theta$ inherits every primitive continuation. The effect family and its primitive actions are declared; complete future behaviour determines the quotient.

> **Theorem CS-1 (translation criterion).** For two such equivalences θ and ϕ, a deterministic common-source translation

$$[m]_\theta \mapsto [m]_\phi$$

exists if and only if θ refines ϕ. When it exists it is unique and commutes with every primitive continuation.

Proof. The displayed assignment is well-defined exactly when $m\theta n$ implies $m\phi n$. Commutation follows because both quotient actions descend from the same source action. □

If neither equivalence refines the other, the framework does not invent a function. It retains the exact same-effect relation

$$\Gamma_{\theta\phi} = \{([m]_\theta, [m]_\phi) : m \in M\}.$$

The relation may be one-to-many in both directions. It is exactly the alignment supported by the common source.

The joint perspective is the common refinement

$$m\theta \wedge \phi n \quad \Leftrightarrow \quad m\theta n \text{ and } m\phi n.$$

Theorem CS-2 (strict pairwise creation). For finite M,

$$\theta \text{ and } \phi \text{ are incomparable} \quad \Leftrightarrow \quad \left|P_{\theta\wedge\phi}\right| > \max\left(\left|P_\theta\right|, \left|P_\phi\right|\right).$$

Proof. If the equivalences are incomparable, each separates a pair merged by the other, so their common refinement strictly splits a class of each. If one refines the other, their meet is the finer equivalence and cannot be larger. □

The common refinement gives a precise meaning to **joint organisation**. It combines the two distinction families and, when they are incomparable, is finer than either marginal quotient. It does not manufacture a pairwise distinction absent from both: if both θ and ϕ identify two source effects, so does $\theta \wedge \phi$. The gain lies in retaining the two inherited families simultaneously, not in altering the source.

Figure 4 displays the three comparison objects supplied by the same source: a direct map under refinement, a relation for incomparable quotients, and their least joint refinement.

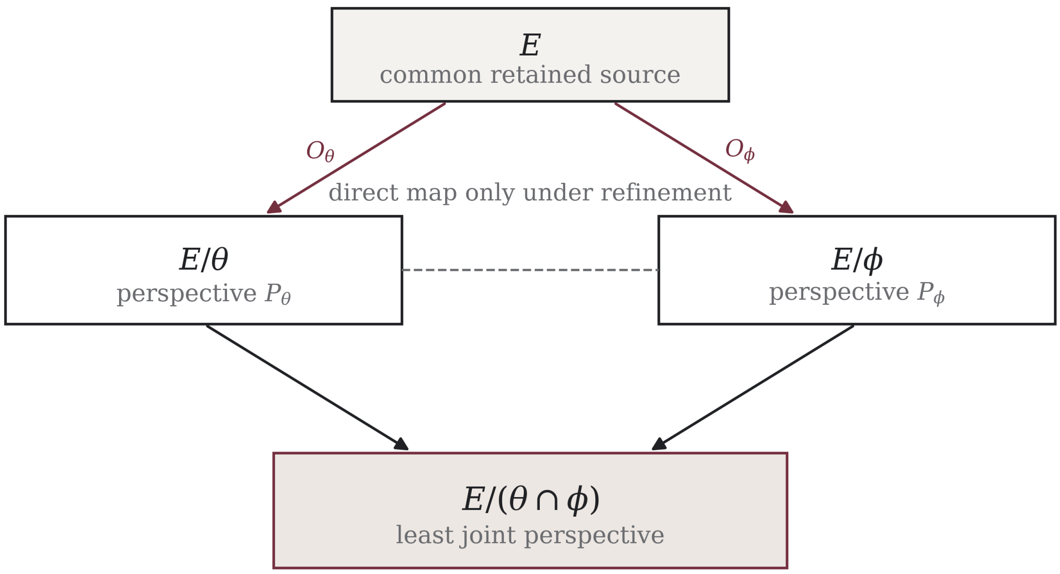

Figure 4. Common-source comparison. Refinement licenses a deterministic translation. Incomparable quotients retain a source-induced relation and meet in a strictly finer joint quotient.

Finite audit. Across the 1,555 canonical source rules in the two-transformation census, the comparison produced 774 equivalent perspective pairs, 3,840 proper one-way pairs, and 51 incomparable pairs. A rule code such as `0012|1130` lists, on each side of `|`, the images of $0, \ldots, n-1$ under one of the two generators. Strict joint growth occurred for all and only the 51 incomparable pairs. Three sources produced a joint calculus larger than each of three audited marginals:

Source rule	**Source effects**	**Kernel**	**Reunion**	**Ideal**	**Joint**
`0012|1130`	13	10	1	8	12

Source rule	Source effects	Kernel	Reunion	Ideal	Joint
`0012|1220`	14	10	1	10	12
`0013|1321`	16	12	1	11	14

Allowing each joint calculus to act as the next source produced closed interaction laws: the first two stabilized as 12-state laws, while the third passed from 14 states to a fixed 11-state law.

Finite underdetermination. Among 376 distinct complete rooted first-perspective laws, 246 admitted multiple compatible source systems. In 20 of those cases the other generated perspective laws differed; in three, even the presence of strict creation differed. These finite pairs refute unique completion within the declared constructor. A complete first-perspective law does not, in this census, select a unique compatible higher-perspective law. The warranted continuation is therefore the full compatible family; selecting the member nearest a desired target would import a new assumption.

Chu, sheaf, and institution formalisms each retain relations or contexts that need not collapse to a single global map (Pratt 1995; Abramsky and Brandenburger 2011; Goguen and Burstall 1992). The present criterion is more specific: both views are first closed against one finite source action, after which refinement decides whether the supported comparison is functional, relational, or a joint quotient.

4.4 A target-free binary distinction game

The preceding constructions begin with effects or laws. This game begins one step earlier. After the first P_1 distinction, any current resultant may distinguish into two retained resultants, and the rule may be applied again. There is no target phenomenon and no global update schedule.

4.4.1 Complete transition family

Declared constructor. Binary continuation is the defining P_1 move. The address marks 0 and 1 prevent accidental identification of occurrences; the transition itself never reads them.

A finite record E is a finite prefix-closed set of binary addresses containing the root ϵ, the first already-present distinction. Its current undistinguished resultants are

$$B(E) = \{w : (\exists v \in E)(w = v0 \vee w = v1),\ w \notin E\}.$$

The next step retains the complete successor family

$$\mathcal{D}(E) = \{(x, E \cup \{x\}) : x \in B(E)\}.$$

The ordered pair retains which existing occurrence was continued and the resulting complete record.

Let σ denote a prefix-preserving automorphism of the rooted binary address tree generated by local exchanges of the marks 0 and 1. It acts componentwise on addresses, records, and transitions:

$$\sigma(x, E) := (\sigma x, \sigma E).$$

Theorem TF (generated continuation laws). The transition family satisfies all of the following.

1. **Local replacement.** For every $x \in B(E)$,

$$B\left(E \cup \{x\}\right) = \{y \in B(E) : y \neq x\} \cup \{x0, x1\}.$$

2. **Inert-name equivariance.** Any recursively applied exchange of the address marks bijects the complete transition family:

$$\sigma\,(\mathcal{D}(E)) = \mathcal{D}(\sigma E).$$

3. **Commutation at separate current occurrences.** If $x \neq y$ lie in $B(E)$, then

$$(E \cup \{x\}) \cup \{y\} = (E \cup \{y\}) \cup \{x\}.$$

4. **Ancestry-only history restriction.** Finite resultant records are exactly the finite prefix-closed address sets containing ϵ. Legal histories of a fixed record are exactly its linear extensions under ancestry. Any two differ by exchanges of adjacent unrelated occurrences.
5. **Common core and common completion.** If E and F are legal, then $E \cap F$ and $E \cup F$ are legal. They obey absorption and distributivity as set identities.
6. **Unique continuation factorisation.** Every extension $F \supseteq E$ decomposes uniquely into one local continuation below each $x \in B(E)$:

$$F_x = \{u : xu \in F\},$$

- where $F_x = \varnothing$ records no continuation below x. Then

$$F = E \cup \bigcup_{x \in B(E)} xF_x.$$

7. **No finite terminal record.** $B(E) \neq \varnothing$ for every finite E.

Proof. If $x \in B(E)$, inserting x removes that boundary address and introduces exactly its two children, which proves local replacement. A prefix-preserving tree automorphism sends boundary addresses to boundary addresses and commutes with adjoining a node; applying this pointwise gives inert-name equivariance. If x and y are distinct boundary addresses, neither is a prefix of the other. Inserting one therefore leaves the other on the boundary, and both update orders produce $E \cup \{x, y\}$.

Every reachable record contains ϵ and is prefix-closed because an address can be inserted only after all of its proper prefixes. Conversely, for a finite prefix-closed set F containing ϵ, order all other elements of F by nondecreasing length. Each address is on the current boundary when selected, so this order constructs F. More generally, a construction history for fixed F is legal exactly when every prefix precedes its descendants—precisely when it is a linear extension of the ancestry order. Any two linear extensions of a finite poset are connected by swaps of adjacent incomparable elements: repeatedly move the next element of one extension leftward through elements that cannot precede it in the other.

Union and intersection preserve prefix closure and the root. For an extension $F \supseteq E$, every new address has a unique longest prefix $x \in B(E)$; the boundary is prefix-incomparable, so the resulting sets xF_x are disjoint and collectively contain precisely the elements of F not already in E. This proves existence and uniqueness of the factorisation. Finally, take an address of maximal length in finite E. Neither child can lie in E, so both are in $B(E)$ and the boundary is nonempty. □

These relations can subsequently be identified with familiar structures. Legal records form a distributive lattice under union and intersection; a fixed record carries an ancestry poset; complete records have rooted full binary-tree shapes; independent local continuations give a recursive species decomposition (Birkhoff 1967; Stanley 2015; Bergeron, Labelle, and Leroux 1998). These identifications describe the result; they do not define the transition.

4.4.2 Enumeration after closure

Counting is applied only after the transition has been fixed. If A_n denotes the number of complete identity records with n distinguished occurrences and $A_0 = 1$ denotes an empty local continuation, unique root factorisation gives

$$A_n = \sum_{i+j=n-1} A_i A_j, \qquad A_n = \frac{1}{n+1}\binom{2n}{n}.$$

The local replacement identity implies that every n-occurrence record has $n+1$ current resultants. The total number of legal histories at level n is consequently $2 \cdot 3 \cdots n = n!$. For a fixed record E, let $s_E(v)$ be the number of distinguished occurrences at or below a non-root occurrence v. Its history multiplicity is

$$H(E) = \frac{(|E|-1)!}{\prod_{v \in E,\ v \neq \epsilon} s_E(v)}.$$

The proof interleaves the independent histories below the two root resultants and proceeds recursively.

Distinguished occurrences	Complete records	Complete legal histories	Current resultants per record
1	1	1	2
2	2	2	3
3	5	6	4
4	14	24	5
5	42	120	6
6	132	720	7
7	429	5,040	8
8	1,430	40,320	9

The finite audit covered 19,631 configuration/boundary identities, 17,576 recursive name exchanges, 67,071 separate-resultant diamonds, 390,625 union/intersection pairs, 6,968 continuation decompositions, and the history formula for all 2,055 records through level eight. No counterexample was found. The all-finite claims rest on the proofs above, not on the bounded audit.

Unlike a cellular automaton, the game has no ambient lattice, neighbourhood, synchronous clock, or selected update table (Wolfram 1984). Its object is an extensible genealogy together with every legal local continuation. The resulting lattice, factorisation, and counting laws are consequences of that transition rather than criteria imposed upon it.

4.5 Projection towers and erased distinctions

A complete record admits several autonomous projections. Studying them together separates closure from mere loss of information.

4.5.1 Four continuation-closed views

Declared projections. Let H_d be the complete set of explicit histories after d distinctions, and consider the tower

$$H_d \xrightarrow{\alpha_d} O_d \xrightarrow{\beta_d} U_d \xrightarrow{\gamma_d} C_d.$$

Here O_d forgets the temporal trace but retains the ordered current form; U_d also forgets every temporary left/right order; and C_d keeps only the number of available occurrences. None is designated as the true view.

For an ordered form T, let d_v be the number of internal occurrences in the subform rooted at v. For an order-free form T, assign each internal occurrence a factor $w_v = 1$ when its two subforms coincide and $w_v = 2$ when they differ, and define

$$e(T) = \prod_v w_v.$$

Then the exact fibre sizes are

$$\left|\alpha_d^{-1}(T)\right| = \frac{d!}{\prod_v d_v}, \qquad \left|\beta_d^{-1}(T)\right| = e(T), \qquad \left|(\beta_d \circ \alpha_d)^{-1}(T)\right| = \frac{d!e(T)}{\prod_v d_v}.$$

The first formula follows by recursively interleaving subtree histories; the second assigns two orientations to every unequal pair of subforms and one to an equal pair. Since every level-d form has $d+1$ available occurrences, C_d has one class and its history fibre has size $d!$.

A view is **continuation-closed** if the multiset of next-view results from all current distinctions depends only on the view-class and not on its representative.

Theorem PT (projection closure and counting barrier). Every view H, O, U, C is continuation-closed. Continued distinction within the count-only view C, however, cannot regenerate any arrangement, orientation, or history distinction erased by $\gamma\beta\alpha$.

Proof. In H_d the complete successor multiset is part of the definition, so closure is immediate. The projection α_d erases only the order in which already recorded distinctions were made; it leaves the current ordered form and its available addresses unchanged. Hence representatives of one O_d-class have identical O_{d+1} successor multisets. Two ordered representatives of one U_d-class differ by finitely many local left/right exchanges. Each exchange bijects their available addresses, commutes with a continuation at the corresponding address, and yields order-free-isomorphic successors. Thus their projected successor multisets agree. Finally, every level-d record has exactly $d+1$ available current occurrences, and every continuation enters the sole class of C_{d+1}. The count successor multiset therefore depends only on d.

For the barrier, induction on continuation length shows that every response generated from a C_d-class remains inside one count class at each later level. An autonomous continuation on C therefore has no response that separates two representatives already merged by $\gamma\beta\alpha$. Arrangement, orientation, and history distinctions cannot be regenerated inside the count-only quotient. □

This is **stable blindness** in a concrete tower. The erased distinctions remain available in H_d, O_d, and U_d, but they are absent from C_d and cannot be recreated by its autonomous continuation. A scalar count can therefore support a simpler law while retaining strictly less source structure.

Figure 5 makes the direction of information loss explicit: every forward projection is continuation-closed, but there is no closed inverse.

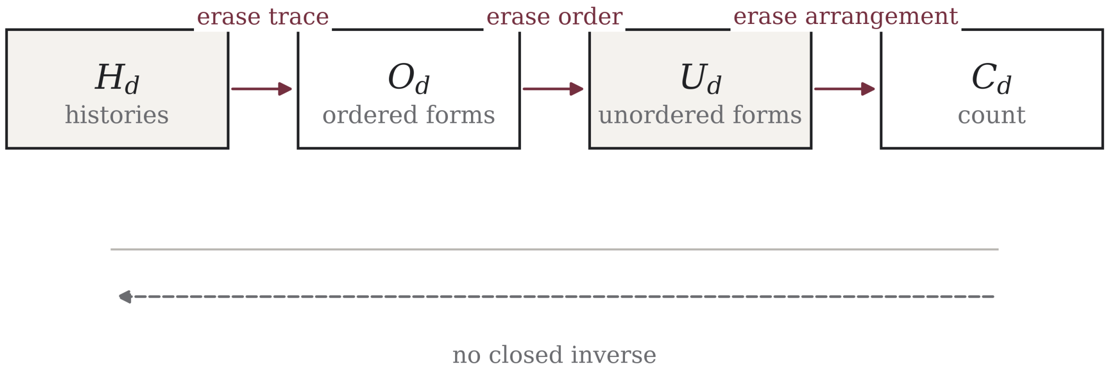

Figure 5. Projection tower. Each quotient supports an autonomous continuation, but no continuation within a quotient recovers distinctions erased by the preceding projection.

4.5.2 4.5.2 Complete partitions before scalar k

Let Π and Σ be continuation-closed partitions of the same forms. Retaining every distinction made by either produces their common refinement

$$x \sim_{\Pi\vee\Sigma} y \quad\Leftrightarrow\quad x \sim_{\Pi} y \text{ and } x \sim_{\Sigma} y.$$

Although the symbol $\vee$ is conventional here, the operation is the intersection of equivalence relations: the coarsest partition refining both. It is forced by complete retention and is commutative, associative, and idempotent.

Let S and T be finite source and target sets, and let $A : S \times T \to \mathbb{N}$ record continuation multiplicity. For a partition Π of T, define the source partition $F_A(\Pi)$ by

$$s \sim_{F_A(\Pi)} s\prime \quad\Leftrightarrow\quad \sum_{t\in B} A(s,t) = \sum_{t\in B} A(s\prime,t) \text{ for every block } B \in \Pi.$$

Theorem PI (partition interaction). If Π refines Σ, then $F_A(\Pi)$ refines $F_A(\Sigma)$. Moreover,

$$F_A(\Pi \vee \Sigma) \text{ refines } F_A(\Pi) \vee F_A(\Sigma).$$

The refinement may be strict.

Proof. Each Σ-block is a union of Π-blocks, so equality of all fine-block counts implies equality of their coarse sums. Joint block counts determine both marginal block-count vectors, proving the second refinement. Marginal vectors need not determine their joint table, so the converse can fail. □

The strict difference defines the interaction-only distinction relation $\mathcal{I}_A(\Pi, \Sigma)$: it consists of source pairs equivalent under $F_A(\Pi) \vee F_A(\Sigma)$ but inequivalent under $F_A(\Pi \vee \Sigma)$.

It contains source pairs that neither marginal pullback separates but the complete joint pullback does. A horizon-eight witness has six classes under separate pullback-and-combination and seven under joint combination-then-pullback. Its two level-six forms

$$((((..).)\,(..))\,(..)) \quad \text{and} \quad (((..)(..))\,((..).))$$

have identical marginal signatures but different joint signatures.

The scalar barrier is stronger than a loss of labels. At level 5, retained partitions with class counts 2 and 3 can interact to produce either 3 or 4 classes. Consequently no function f satisfies

$$|\Pi \vee \Sigma| = f(|\Pi|, |\Sigma|)$$

even in this finite generated family. Before a further projection is specified, the complete objects are the partitions themselves,

$$K_P = \Pi_P, \qquad I(P, Q) = \mathcal{I}_A\left(\Pi_P, \Pi_Q\right),$$

not their cardinalities. A numerical k is a downstream shadow. Information measures based on scalar costs over a fixed alphabet remain valid in their intended domains (Hartley 1928; Shannon 1948; Cover and Thomas 2006); they begin after the distinctions kept explicit here have already been fixed.

4.6 Conditional laboratory: an open finite-rule census

The target-free game isolates the consequences of one binary continuation. A separate laboratory admits richer source structure in order to study the closed laws and perspectives it generates. Its findings are conditional on that choice of source.

4.6.1 Declared branch

Choose a finite carrier $X = \{0, \dots, n-1\}$, with $n \in \{2, 3, 4\}$; two arbitrary total transformations $a, b : X \to X$; identity; and ordinary composition. Products are written $gm := g \circ m$, so m acts first. Retain every unordered primitive pair with repetition, quotienting only by carrier relabelling and exchange of primitive names. These are declared laboratory conditions, not consequences of the sole distinction rule. Algebraic claims below hold for every finite system satisfying their displayed hypotheses. Counts and classifications are exhaustive only at the stated carrier sizes.

The exhaustive finite census begins with the following source counts.

Carrier size	Raw primitive pairs	Canonical systems	Distinct invariant profiles
2	10	7	6
3	378	74	67
4	32,896	1,474	1,175

Within this census, two primitive transformations generate up to 176 effective transformations.

> **Proposition OR-1 (ideal and rank order).** In every generated transformation monoid M, composition induces a one-way preorder. For the two-sided ideal $I(m) = MmM$ and every $g \in M$,

$$rank(gm) \le rank(m), \qquad I(gm) \subseteq I(m).$$

Proof. The image of gm is the image under g of the image of m, so it has at most $rank(m)$ elements. Moreover,

$$I(gm) = MgmM \subseteq MmM = I(m),$$

because $Mg \subseteq M$. Repeating the argument after any further left composition proves that a strict ideal descent cannot later be reversed. □

Further composition cannot reverse strict ideal descent, although recurrence may occur within a level. This is an internal order and carries no automatic interpretation as time, entropy, causality, or knowledge.

4.6.2 Quotients and predictive closure

Among the 1,474 four-state systems, 1,171 have proper nontrivial congruence quotients, 529 have incomparable quotient pairs, and 270 exhibit distributed effect recovery: no proper quotient is faithful alone, while the retained quotient family is jointly faithful. For rule `1002|2210`, two incomparable three-block quotients each distinguish 7 of 14 generated effects and together distinguish all 14.

> **Proposition OR-2 (kernel and predictor order).** Each generated effect m has a kernel κ_m, where $x\kappa_m y$ iff $m(x) = m(y)$. Along $m \mapsto g \circ m$, kernels can only coarsen. If K is the number of current-kernel classes, Q the number of classes under

$$m \equiv_\kappa n \quad \Leftrightarrow \quad \ker(xm) = \ker(xn) \text{ for every } x \in M,$$

and $E = |M|$ the number of full effects, then

$$K \leq Q \leq E.$$

Proof. If $m(u) = m(v)$, then $g\,(m(u)) = g\,(m(v))$, so $\kappa_m \subseteq \kappa_{gm}$. For the class count, equality of effects implies $\equiv_\kappa$, while $m \equiv_\kappa n$ implies $\ker(m) = \ker(n)$ by taking x to be the identity. Thus full-effect equality refines $\equiv_\kappa$, which refines current-kernel equality, and the displayed inequality follows. □

The present kernel often fails to determine its successor. Among four-state systems, 1,446 show strict kernel descent, 811 have incomparable successor-kernel branches, 995 show kernel-level predictive failure, and 1,128 reach kernels that are not global congruences.

The equivalence $\equiv_\kappa$ is therefore the predictive quotient: it retains exactly the distinctions required to determine every future kernel history.

The four-state systems divide into three predictive regimes:

Four-state regime	Systems	Interpretation inside the branch
$Q = K$	479	current kernel is already autonomous
$K < Q = E$	158	every full-effect distinction is required
$K < Q < E$	837	a genuine intermediate predictor is forced

For `0032|2031`, $K = 12$, $Q = 137$, and $E = 176$: 39 full-effect distinctions are irrelevant to every future kernel history, but the kernel alone omits distinctions needed to predict its future.

Theorem OR-3 (recursive-lift termination). Treating the predictor as a transformation system and repeating the construction gives a finite recursive lift. After the first lift each proper step is a strict quotient. Up to pointed-monoid isomorphism, every finite trajectory reaches a fixed point and cannot enter a nontrivial cycle.

Proof. The relation $\equiv_\kappa$ is a two-sided monoid congruence. If $m \equiv_\kappa n$, then for any $y \in M$ and every left context x,

$$\ker(xym) = \ker(xyn),$$

which proves left stability. For right stability, equality of $\ker(xm)$ and $\ker(xn)$ implies equality after pullback by y:

$$\ker((xm)y) = y^{-1}(\ker(xm)) = y^{-1}(\ker(xn)) = \ker((xn)y).$$

Hence the predictive classes form the quotient monoid $C = M/{\equiv_\kappa}$. The lifted system is the left-regular action of C on itself, $[u] : [m] \mapsto [um]$. Distinct elements induce distinct transformations because their actions on the identity class differ, so the lifted carrier and its generated effect monoid both have cardinality $|C|$. Identifying C with this faithful regular representation, every subsequent lift is a quotient of the preceding finite monoid. A proper lift strictly reduces cardinality; a nontrivial cycle would require a later increase or a proper quotient of equal cardinality, both impossible. Finite descent terminates at a fixed point up to pointed isomorphism. □

Across all 1,555 canonical systems, 1,523 are fixed after the first lift, 32 require one further quotient, and all converge to 369 pointed fixed systems. The largest expansion is $4 \to 137 \to 137$; the largest second reduction is $4 \to 8 \to 4 \to 4$.

4.6.3 Endogenous futures and irreversible opportunity loss

Proposition OR-4 (future-domain and reunion order). Let a finite monoid M act on X. For $x \in X$, put $I_x = Mx$. Every $g \in M$ satisfies $I_{gx} \subseteq I_x$, and $I_x = I_y$ exactly when x and y are mutually reachable.

Proof. Since $Mg \subseteq M$, we have $I_{gx} = M(gx) = (Mg)x \subseteq Mx = I_x$. If $I_x = I_y$, then $x \in I_y$ and $y \in I_x$, so each state reaches the other. Conversely, mutual reachability gives both inclusions between the generated domains. □

Across the 369 fixed systems, 11,211 states generate 3,059 distinct principal domains; 368 systems have strict domain descent, 246 have incomparable primitive branches, and 29 have multiple minimal terminal domains.

Reunion-capacity monotonicity. For an unordered pair x, y, define reunion capacity by three exhaustive cases:

- $\chi(x, y) = 2$ when some $u \in M$ satisfies $u(x) = u(y)$;
- $\chi(x, y) = 1$ when $I_x \cap I_y \neq \varnothing$ but no common u merges the pair;
- $\chi(x, y) = 0$ when $I_x \cap I_y = \varnothing$.

Then every generated continuation obeys

$$\chi(gx, gy) \le \chi(x, y).$$

Proof. If the target pair (gx, gy) has a merger u, then ug merges (x, y), so capacity 2 at the target implies capacity 2 at the source. If $I_{gx} \cap I_{gy}$ is nonempty, the domain inclusions already proved give $I_x \cap I_y \neq \varnothing$;

thus positive target capacity implies positive source capacity. These are all three ordered cases, proving the inequality. □

Opportunity may therefore be lost but cannot be recovered under a shared continuation.

The complete census classifies 347,335 unordered distinct pairs: 346,991 are coalescible by a common continuation, 131 overlap only under different continuations, and 213 have disjoint future domains. It records 347 strict primitive decreases: 37 of type $2 \to 1$, 207 of type $2 \to 0$, and 103 of type $1 \to 0$. The monotonicity inequality was independently checked in 31,930,235 pair/effect applications.

For an effect m, the future-profile construction retains $\Lambda_m(x, y) = \chi\,(m(x), m(y))$ and identifies two effects exactly when all their future profiles agree.

Present reunion capacity is not always autonomous. Across the 369 fixed systems, 11,211 effects yield 484 current profiles, while complete future prediction requires 552 states. Twenty-two systems require hidden refinement, and four force an intermediate carrier strictly between present-profile and full-effect descriptions. Recursive future-profile closure terminates in 22 fixed systems, with no expansion or nontrivial cycle.

4.6.4 What the laboratory establishes—and what it cannot establish

The census supplies exact examples of transformation growth, irreversible ideal and kernel orders, incomparable quotients, distributed recovery, minimal predictors, recursive closure, distinct local futures, monotone opportunity loss, and interaction-created distinctions. It also gives counterexamples to several scalar and functional simplifications.

The finite carrier, total transformations, composition, identity, and two-primitive source are laboratory inputs. They are not derived from P_0/R or from the binary constructor. Kernel depth is not equated with k, a predictive class with an actual perspective, or ideal descent with physical time. Exhaustion at $n \leq 4$ establishes the census, not a theorem beyond it.

The laboratory separates two questions: what follows from the declared interaction, and whether that interaction models a named phenomenon. Only the first is addressed here. Fifty-six automated tests cover all 1,555 canonical systems, 32,926 generated effects, 28,313 ordered quotient pairs, 20,317 minimised predictive states, 31,930,235 opportunity-loss checks, and an independent reconstruction of every recorded quotient and profile. These tests assess the calculus and its implementation; they do not privilege one audited system as a model of reality.

The falsifiability section records the corresponding counterexamples and provenance failures. None of the finite representations is asserted to exhaust the possible perspectives of R.

5 Prime-source occurrences and the 36-relation frontier

The elementary identity $p \cdot 1 = p$ supplies a compact example of the source/report distinction. The two p tokens have equal value but different source and returned-product roles. A value reporter merges them; an occurrence-sensitive reporter retains the relation

$$p_{\text{source}} \cdot 1_{\text{unit}} = p_{\text{return}}. \tag{1}$$

The reduced triad contains every positive factor relation exactly when p is prime. For a composite n, an intermediate factor pair is a separator that the triad omitted.

The finite terminal carrier $\mathbb{Z}_{10}$ gives a second example. Removing the zero terminal leaves nine active states. The complete unordered relation frontier contains $\binom{9}{2} = 36$ cells. Under the affine return $d \mapsto 1 - d \pmod{10}$, the eight interior states form the four pairs $(2, 9)$, $(3, 8)$, $(5, 6)$, and $(7, 4)$. A reporter that keeps only the prime-sector label identifies the two sides of each pair. It is closed only for continuation families that cannot distinguish source from return.

Proposition 5.1 (Prime-observer quotient test)**.** *Let $U(q, \varepsilon) = q$ on the four two-sided sectors. An autonomous law descends to the U-states if and only if every admitted continuation has the same U-response on $(q, +)$ and $(q, -)$ for each $q \in \{2, 3, 5, 7\}$.*

Proof. This is the autonomy/closure criterion applied to the side-forgetting fibres. A distinguishing continuation witnesses failure; fibrewise constancy defines the quotient law. □

The example connects the abstract fibre condition to an arithmetic occurrence: equality of present value is insufficient whenever role, multiplicity, or source address changes a future operation. P17 gives the complete $8 + 4 + 24$ relation decomposition and the all-prime predictive refinement.

6 Related work

Spencer-Brown's *Laws of Form* gives distinction formal priority through the mark and the calculus of indications (Spencer-Brown 1969). Varela develops related calculi of self-reference and autonomy (Varela 1975, 1979). The present construction shares the foundational emphasis on distinction but takes a different mathematical route: the primitive report is an unordered complete family with occurrence multiplicity, and the central objects are reconstructive views, complete output decks, and quotient reporters. No crossing law, re-entry rule, or Boolean reading is assumed.

The language of perspective and autonomy also places the work near second-order cybernetics, autopoiesis, and the law of requisite variety (Foerster 1981; Maturana and Varela 1980; Ashby 1956). Conant and Ashby's good-regulator theorem makes the relation between regulation and internal modelling particularly explicit (Conant and Ashby 1970). Here, however, perspective has a fixed formal meaning: it is a quotient of a declared source. Autonomy is the descent of complete behaviour through that quotient, and its associated limit is the stable-blindness theorem.

The core calculus is prior to probability. The discrete coding formulations of Hartley and Shannon begin with a specified alphabet and a quantitative coding or distributional problem (Hartley 1928; Shannon 1948; Cover and Thomas 2006). Complete decks instead retain occurrence multiplicity before normalisation or randomisation. Blackwell's comparison of experiments and the data-processing behaviour of Kullback–Leibler divergence provide useful analogies for information loss under post-processing (Blackwell 1953; Kullback and Leibler 1951), but the results here are deterministic statements about quotient fibres. No probability measure or divergence enters their proof.

The closest technical neighbours of response closure are behavioural equivalence and partition refinement. Myhill–Nerode theory distinguishes states by their future contexts (Myhill 1957; Nerode 1958; Rabin and Scott 1959); classical algorithms compute stable partitions efficiently (Moore 1956; Hopcroft 1971; Paige and Tarjan 1987). Coalgebra provides a general language for observations, transitions, and behavioural equivalence (Rutten 2000; Wissmann et al. 2020), while universal algebra treats quotients through operation-preserving congruences (Birkhoff 1935; Burris and Sankappanavar 1981). Transformation semigroups supply a related setting for finitely generated effects (Eilenberg 1976; Howie 1995). The distinction-generated setting contributes a particular provenance: complete occurrence decks are derived from the source hierarchy, and disagreement between those decks is used to generate—not merely test—the next refinement.

The hierarchy itself lies in the combinatorics of rooted trees, species, and recursive classes (Otter 1948; Joyal 1981; Bergeron, Labelle, and Leroux 1998). Classical reconstruction asks, among other variants, whether a structure can be recovered from deletion cards (Kelly 1957). The deck used here is different: each retained occurrence is replaced once, and origin multiplicity is preserved. The target-free binary process produces Catalan counts (Stanley 2015), but the enumeration is observed only after the transition has been fixed. This reverses the usual modelling order in cellular automata, where a lattice, neighbourhood, alphabet, and update rule are selected before emergent behaviour is studied (Wolfram 1984; Crutchfield 1994).

Common-source comparison can be located more precisely among categorical and contextual formalisms. Institutions formalise translation across signatures (Goguen and Burstall 1992); Chu spaces retain an explicit object–observation pairing (Pratt 1995); and sheaf- and topos-based approaches encode compatibility among contextual descriptions (Abramsky and Brandenburger 2011; Döring and Isham 2008). The present result is narrower: two quotient maps from a single finite source determine whether their comparison is a function, a genuinely relational span, or a joint refinement. The order-theoretic part belongs to the lattice of equivalence relations, with intersection as meet (Birkhoff 1967), while its interpretation as a least shared report parallels the intent, though not the full machinery, of formal concept analysis (Ganter and Wille 1999).

6.1 Position of the contribution

None of the individual ingredients—rooted hierarchies, multisets, partitions, congruences, closure operators, behavioural equivalence, or Catalan enumeration—is new. The contribution claimed here is a source-closed theorem chain: complete recursive distinction yields a source hierarchy; that hierarchy determines complete behaviour; and complete behaviour determines either a lawful quotient or the least forced refinement.

Its main assertion is a coupled one: constancy of complete behaviour on quotient fibres is exactly the condition for autonomous law; failure determines a least missing distinction; and success fixes an internal ceiling on reconstruction. The one-occurrence reconstruction theorem, the relational obstruction to naive reporter pairing, and the common-source treatment of incomparable reports complete that chain. The novelty claim is deliberately synthetic rather than absolute: rooted hierarchies, congruence refinement, behavioural equivalence, and partition lattices are established mathematics. What is claimed here is their integration under one source-retention discipline, together with the stated reconstruction, least-closure, stable-loss, and reporter-combination results. Any stronger priority claim would require an equivalence analysis preserving occurrence multiplicity, reporter order, and the relevant universal properties.

7 Limitations and falsifiability

The boundary P_0/R has no positive theory here. Assigning it even an empty carrier would place a distinction at the point reserved for no reportable distinction. Likewise, when $q > 2$, a q-resultant record in P_1 supports theorems about that record but says nothing by itself about the native logic or experience of a hypothetical P_{q-1}.

The hierarchy grammar is minimal only relative to stated retention requirements: a containing occurrence, all present resultants, multiplicity, recursion, and no unlicensed child order. Other faithful encodings may exist. The proved reconstruction results are finite and well-founded; infinite hierarchies, non-well-founded recurrence, limiting closure, measurability, and effective computation require additional hypotheses. Non-well-founded set theory may supply tools for one such extension (Aczel 1988), but it is not part of the present core.

The finite interaction branches deliberately add carriers, operations, focus values, effect families, or schedules. Those assumptions appear in the relevant theorem statements and are marked [X] in Appendix A. Formal resemblance to time, space, causality, energy, learning, or another empirical concept is insufficient for identification. Such a bridge would require a measurement map and discriminating predictions.

The mathematical claims are directly refutable. Counterexamples would include unequal finite hierarchies with equal reconstructive views at the stipulated depth; unequal sources with the same complete one-occurrence deck; a reporter satisfying fibre constancy but lacking a descended law; a closed refinement properly below the claimed least closure; or an already-closed reporter whose response iteration refines its fibres. The finite audits are refuted by an independently reproducible discrepancy in domain, canonicalisation, totals, or witnesses.

The provenance conditions are also testable. A [B] claim is violated if a proof later assigns structure to P_0/R; a [C] construction is defective if it silently introduces order, labels, selection, probability, deletion, or a target; an [X] result is mis-scoped if its added assumptions disappear from the theorem statement. Beyond formal correctness, the programme would have limited scientific value if its bridges proved uninformative, its complete decks intractable, or its theorem chain reducible to a known formalism without a useful gain in provenance or application.

8 Reproducibility and research artefacts

Proofs carry the mathematical claims. Code is used to audit finite domains, retain counterexample witnesses, and expose assumptions that may be hidden in an implementation. The companion archive, *Abhijit_Singh_Distinction-Generated_Perspectives_v2.0_source.zip*, contains the manuscript sources, figures, cited bibliography, theorem-oriented test corpus, complete finite-audit data, and a single audit runner. Its *REPRODUCIBILITY.md* records the file manifest, Python version, exact invocations, and expected totals.

Running *bash run_audits.sh* from the archive root executes the curated suite under Python 3.12. The suite contains **134 automated test methods**: 21 core hierarchy-and-closure tests, 38 same-referent, predictive, and universal-law tests, 7 target-free-game tests, 12 projection-and-partition tests, and 56 open-rule-census tests. The largest enumerations cover all 59,049 pointed binary-input ternary laws and 1,555 canonical finite source rules in the common-source study. The release documentation supplies the canonicalisation rules, test modules, expected totals, and machine-readable witnesses; the bounded computations remain logically separate from the unbounded proofs.

These computations support diagnosis and independent replication; they do not replace unbounded proofs or establish the boundary and constructor choices.

9 Open problems

The following problems arise directly from the present calculus.

1. **Complete two-distinction interaction.** Classify the full two-step families for identical, nested, and incomparable selected occurrences without assuming a cut. Determine when opposite orders agree, refine one another, or force a relational report.
2. **Infinite and non-well-founded completion.** Identify infinite classes on which finite views determine a source and least response closure exists; state the topological or coalgebraic assumptions explicitly.
3. **Encoding invariance.** Characterise the theorems preserved by any faithful P_1 encoding of incidence and multiplicity.
4. **Complexity of forced refinement.** Obtain sharp bounds for computing $Cl_\infty(Q)$, and locate tractable subclasses relative to classical partition-refinement algorithms.
5. **Composition of perspectives.** Develop a categorical setting for closed reporters, source-compatible translations, and relations, treating the common-source relation as primitive when no translation exists.
6. **Empirical bridge criteria.** State necessary conditions for identifying a derived structure with an observed regularity, including behaviour preservation, risky predictions, and instrument-induced erasure.
7. **Nonbinary representation.** For $q > 2$, determine the strongest encoding-invariant statements P_1 can make about a complete q-resultant presentation without claiming native P_{q-1} access.

10 Conclusion

The paper begins at a carefully delimited point: P_0/R has no positive model, while P_1 can record complete resultant families, recursive containment, and occurrence multiplicity. That limited transcription already supports two reconstruction theorems and a general criterion for the descent of law to a quotient reporter.

The criterion links autonomy to loss. A quotient is lawful precisely when the distinctions it erases do not alter complete reported response. If that condition fails, behaviour identifies the least required refinement. If it holds, the descended law cannot recover the erased distinctions. Closed reporter combination and common-source comparison show that relational information may become necessary again when perspectives interact.

This is a finite formal theory, not an empirical ontology. Its contribution is an explicit calculus in which source construction, quotient law, forced refinement, and stable blindness belong to one theorem chain. Further claims must state their added structure and earn any interpretation through a separate bridge.

11 References

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12 Appendix A. Theorem and provenance register

The following register records the principal claims and the strongest defensible status assigned to each. Codes are **[B]** boundary discipline, **[C]** declared constructor, **[G]** general theorem, **[F]** finite audit, and **[X]** conditional extension. Here "general" is restricted to the declared mathematical domain and does not extend to the non-reportable boundary.

ID · status	**Result**	**Statement and scope**
A1 [B]	P_0/R boundary	No carrier, object, state, law, value, or operation is assigned to the absence of reportable distinction.
A2 [B]	First reportable perspective	Every symbol and computation is a binary report in the first reportable perspective; a nonbinary q-resultant presentation with $q > 2$ is a P_1 encoding, not native access to the corresponding nonbinary perspective.
A3 [C]	Finite hierarchy grammar	The unordered-multiset grammar with terminal • and internal q-resultant nodes, $q \geq 2$, is the minimal declared no-discard transcription.
A4 [C] [G]	Prop. 3.2.3: persistent occurrence	Under A3's declared no-erasure, no-selection, and no-duplication obligations, the container uniquely retains one old role through distinction.
A5 [C]	Present-resultant operator	Δ returns the complete multiset of already-present child reports; no child is selected or weighted.
A6 [G]	Thms. 3.3.2–3: view reconstruction	For height $h(T)$, the depth-$h(T) + 1$ recursive complete-resultant view determines T; view depth strictly refines.
A7 [G]	Prop. 3.2.4: balance	$L(T) = 1 + \sum_v (q(v) - 1)$ for every finite hierarchy.
A8 [G]	Lossy scalarization	Terminal count, internal count, and flattened profiles identify explicit pairs of unequal hierarchies. Nested relation cannot be reconstructed from their scalar values.
A9 [G]	Thm. 3.4.2: deck reconstruction	For fixed reported $q \geq 2$, equality of complete one-occurrence decks $E_q(T) = E_q(U)$ implies $T = U$.

ID · status	Result	Statement and scope
A10 [G]	Thm. 3.4.3: cut and reunion	For a marked occurrence $T = C[D]$, internal and external one-step outputs partition the full deck exactly, with occurrence origin and multiplicity retained.
A11 [G]	Thm. 3.4.4: restricted order equality	Replacements at disjoint occurrences on opposite sides of a persistent cut preserve one another and produce the same complete two-step family in either order.
A12 [G]	Thm. 3.5.2: quotient closure	A quotient law exists exactly when equal Q-reports have equal complete pushed-forward output decks.
A13 [G]	Thm. 3.6.1: least response closure	Iterated complete response reports define $Cl_\infty(Q)$, the least closed refinement of Q relative to the declared output decks.
A14 [G]	Thm. 3.7.1: stable blindness	If Q is closed, every response iteration has the same fibres as Q; its own law cannot recover stably erased distinctions.
A15 [G]	Prop. 3.7.2: grouping hierarchy	Closed reporters $G_0 \prec G_1 \prec \cdots \prec G_\omega \simeq \mathcal{H}$ retain successively deeper shared containment; adjacent laws commute with erasure.
A16 [G]	Thm. 3.8.1: closed combination	The response closure of the direct pair defines $X \vee_c Y$, the least reporter above both that supports the complete law. Cor. 3.8.2 gives its commutative, associative, and conditional idempotence laws.
A17 [C] [G]	TF: target-free binary game	Prefix-closed binary records with all legal boundary expansions retained satisfy local replacement, inert-name invariance, separate-occurrence commutation, ancestry-only history restriction, exact factorisation, and no finite terminal state.
A18 [F]	Binary-game enumerations	Complete-state counts for sizes 1–8 are 1, 2, 5, 14, 42, 132, 429, and 1,430; complete legal histories are $1!, \ldots, 8!$, with finite identity and decomposition audits.

ID · status	Result	Statement and scope
A19 [C] [G] [F]	PT/PI: projection and partition closure	Explicit histories project to ordered forms, order-free forms, and availability; each projection is continuation-closed, and partition interaction is monotone under refinement. No count-only reporter recovers erased form.
A20 [X] [G]	SRQ: same-referent quotient	For a declared effect family A, equality under all effects defines the least sufficient quotient Q_A; a candidate descends iff constant on its fibres.
A21 [X] [G]	CR: constraint reversal	If Q_2 properly refines Q_1, then for every codomain with at least two values the family of compatible candidate laws is strictly larger over Q_2.
A22 [X] [G]	RLD: completion count	For a finite codomain of size $q \geq 2$, a reached partial law with r reached configurations among M admits exactly q^{M-r} total completions; uniqueness holds exactly when $r = M$. For $q = 1$, the extension is uniquely constant regardless of reachability.
A23 [C] [X] [G]	IPQ: predictive quotient	For a finite carrier $\lvert X \rvert = q \geq 2$ and arity $m \geq 1$, finite one-place contexts of a pointed operation generate a stabilizing congruence; its quotient is the coarsest focus-preserving predictive quotient.
A24 [F]	Predictive audit	The finite ternary audit checks all 59,049 pointed binary-input laws, including the stabilisation bound and quotient universal property.
A25 [X] [G]	CS-1: common-source translation	For quotient perspectives of one declared finite source, a direct translation exists exactly in the refinement direction; otherwise the full source-induced relation is retained.
A26 [X] [G]	CS-2: joint perspective creation	Incomparable source quotients induce a joint quotient strictly finer than either when each retains a distinction erased by the other.

ID · status	Result	Statement and scope
A27 [F]	Common-source audit	The translation, relation, and joint-perspective claims are checked across all 1,555 frozen finite source rules in the declared audit domain.
A28 [C] [X]	Open finite-rule constructor	The laboratory declares a carrier of size $n \in \{2, 3, 4\}$, two total transformations, identity, and composition; none is derived from the primitive.
A29 [X] [G]	Prop. OR-1: ideal and rank order	In the declared transformation monoid, $rank(gm) \le rank(m)$, while multiplication can only contract the generated two-sided ideal.
A30 [X] [G]	Prop. OR-2: kernel and predictor order	Kernels can only coarsen under left composition, and current-kernel, minimal-predictor, and full-effect counts obey $K \le Q \le E$.
A31 [X] [G]	Thm. OR-3: recursive predictive lift	Equality of every future kernel history is a two-sided congruence. Up to pointed-monoid isomorphism, the induced finite lift reaches a fixed point and has no nontrivial cycle.
A32 [X] [G]	Prop. OR-4: future-domain and reunion order	Generated domains satisfy $I_{gx} \subseteq I_x$, and reunion capacity satisfies $\chi(gx, gy) \le \chi(x, y)$ under every shared continuation.
A33 [F]	Open-rule census	Exhaustive enumeration for $n \le 4$ records 1,555 canonical systems, their quotient and predictor regimes, recursive lifts, reunion profiles, and all stated opportunity-loss audits.